L06 — study notes

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Lecture 6 — Repeated games II: collusion and relational contracts · Map of the notes

Estimated unit. Built on 2026-10-16 from the study guide before the lecture. Reconcile with the slides when Aula/Lecture06.pdf arrives (/notes 06 –reconcile).

One claim. A promise nobody can enforce in court can still be kept, if breaking it costs more than it pays. In an infinitely repeated game the gain from cheating today is weighed against the discounted loss of the relationship, and cooperation is self-enforcing iff $\delta \geq (v^d - v^c)/(v^d - v^p)$. **Firms use it to collude without a contract** — 9/17 for a Cournot duopoly, $1 - 1/n$ for n Bertrand firms. **Traders use it to be trusted** — a reputation is that inequality held in front of many partners, and institutions lend it to strangers. **Firms use it to buy effort** — an efficiency wage is the rent that makes the job worth keeping honestly.

Three layers per topic:

Layer	What it means here
U — intuition	Gibbons's applied prose, M&R's cases (Lord Clive, Maghribi traders, trench truces, MacRents), Cabral's industry boxes
C — course (examinable)	Tadelis (10.5) and its applications in §10.4–10.5; Gibbons's (2.3.2)–(2.3.7) with every derivation; Cabral (9.1)–(9.5); M&R (8.1) and the trust box
G — graduate	MWG Prop. 12.D.2's "nothing below 1/2" proof; imperfect monitoring (Green–Porter, Abreu–Pearce–Stacchetti); renegotiation-proofness. **Named, not examinable**

Reading order

Part I — the tool

1. L06-01-recap-grim-trigger — Tadelis §10.2–10.3 (pp. 192–200), recap only: the one-stage deviation principle, grim trigger, and **(10.5) derived step by step** in its three forms (cutoff, temptation-vs-stake, average payoff); the PD's $\delta \geq 1/4$.

Part II — collusion

2. L06-02-cournot-collusion — Tadelis §10.4 (pp. 201–203), Gibbons §2.3.C (pp. 102–104): the 22/23 split with **a corrected firm-2 cutoff (0.473, not the printed 0.392)**; the symmetric 9/17; the **shrinking cartel** $q^*(\delta)$; the n-firm cutoff $(n+1)^2/(n^2+6n+1)$.
3. L06-03-bertrand-market-structure — Cabral §9.1, §9.3 (pp. 218–231), MWG §12.D: $\delta \geq \frac{1}{2}$ and $1-1/n$, **all or nothing**; Cabral's (9.5) discount factor and the **121-firm** cartel; few, similar firms and multimarket contact.
4. L06-04-punishments-price-wars — Gibbons pp. 104–107, Tadelis p. 203, Cabral §9.2: Abreu's **carrot and stick** at $\delta = 1/2$, $x/(a-c) \in [3/8, 1/2]$; renegotiation; **price wars on the equilibrium path** under noisy monitoring; counter-cyclical collusion.

Part III — reputation

5. L06-05-reputation-trust — Tadelis §10.5.1 (pp. 204–205), M&R pp. 259–266: the trust game's $\delta \geq 1/2$; M&R's box and multiplier $N = 1/(1-\delta)$, with **the exact** $NV \geq V + G$; reputation as a series of partners; live and let live; the limits.

6. L06-06-institutions-reputation — Tadelis §10.5.2–10.5.3 (pp. 205–209), M&R pp. 266–268: the guarantor's $\delta \geq 0.95$, the law merchant, brand names as bonds, the Hansa.

Part IV — employment

7. L06-07-efficiency-wages — Gibbons §2.3.D (pp. 107–112), M&R pp. 250–255: **(2.3.5) derived** from V_e and V_s; the firm's $\delta_{\min}$; M&R's USD 60,000 efficiency wage and **the mapping between the two models** (two opposite p's).
8. L06-08-relational-contracts — synthesis: **one inequality, six relationships**, the design levers, and when relational contracts fail.

Support

9. L06-09-formulas — every result and number; the study-guide block mapped item by item.
10. L06-10-pitfalls — twelve confusions and 24 original True/False items.
11. L06-11-python — every routine with `assert` checks against the books' numbers.

The one result to hold

$$\underbrace{v^d - v^c}_{\text{gain from cheating today}} \leq \frac{\delta}{1-\delta} \cdot \Pr(\text{caught}) \cdot \underbrace{(v^c - v^p)}_{\text{what the relationship is worth per period}}$$

Every relational contract in the course is this line. A cartel member, a trusted merchant, a guarantor and a well-paid worker differ only in what v^c, v^d, v^p and the detection probability are.

Numbers to have memorized

Case	Cutoff / value	Source
Repeated PD (4, 5, 1)	$\delta \geq 1/4$	Tadelis p. 200
Cournot duopoly, half monopoly each	$\delta \geq 9/17 = 0.529$	Gibbons p. 103
Tadelis's 22/23 split	0.595 (firm 1), **0.473** (firm 2)	p. 202, corrected
Bertrand, n firms	$1 - 1/n$: 0.5, 0.667, 0.75	Cabral (9.4); MWG 12.D.1
Monthly prices at 10% a year	$\delta = 0.9917$, up to 121 firms	Cabral p. 221
Carrot and stick at $\delta = 1/2$	$3/8 \leq x/(a-c) \leq 1/2$	Gibbons p. 106
Trust game / guarantor	1/2 / 0.95	Tadelis pp. 205, 207
Efficiency wage, M&R	USD 60,000 ($N = 1$), 50,000 ($N = 2$)	M&R pp. 251–252

Figures

Built by `figures/make_figures.py` (run: `python Notas/L06/figures/make_figures.py`; it ends with `assert` checks on the plotted numbers).

Figure	Shows	Lives in
L06-cournot-cutoffs	conform vs deviate for the 22/23 split; cutoffs 0.595 and 0.473, printed 0.392 marked	L06-02-cournot-collusion §2
L06-best-sustainable-q	Gibbons's $q^*(\delta)$ and the profit it delivers	L06-02-cournot-collusion §4
L06-threshold-vs-n	Bertrand vs Cournot cutoffs by n; largest cartel by pricing frequency, Cabral (9.5)	L06-03-bertrand-market-structure §2
L06-carrot-stick	slack of (2.3.3) and (2.3.4) at $\delta = 1/2$; the band $[3/8, 1/2]$	L06-04-punishments-price-wars §1
L06-trust-game	Tadelis Fig. 10.1 as a tree; thresholds 1/2 and 0.95	L06-05-reputation-trust §1
L06-efficiency-wage	$w^*(\delta)$ at M&R's calibration for three detection rates	L06-07-efficiency-wages §6

Interactive companions

- The Cartel Patience Dial (`companions/companion-collusion-threshold.html`) — Cournot or Bertrand, n, δ, a, $c \to$ the cutoff δ^* (the answer), collusive, deviation and punishment profits, the most collusive sustainable output and price, and the cutoffs by n for both modes; "snap to δ^*" lands on the landmark. Linked from notes 02 and 03.
- The No-Shirking Wage (`companions/companion-efficiency-wage.html`) — δ, p, e, w_0, $y \to w^*$ (the answer), the rent, M&R's N, the firm's margin and $\delta_{\min}$, and the effort/shirk value lines crossing at w^*. Defaults reproduce M&R's USD 60,000. Linked from note 07.

Both are verified by `companions/check_companions.py` (Python vs the pages' own JavaScript under Node).

Scope and ceiling

- **Examinable (estimated):** (10.5) and its derivation; tacit collusion in Cournot and Bertrand, with n firms; the factors in Cabral (9.5) and §9.3; Abreu's two-phase strategy at the level of Gibbons; reputation in the trust game and M&R's box; the guarantor; Gibbons's efficiency-wage model and M&R's (8.1).
- **Named only:** imperfect monitoring beyond the intuition (Green–Porter, APS); MWG's proof that nothing survives below $\frac{1}{2}$; renegotiation-proofness; reputation as beliefs about types (Tadelis Ch. 17, Gibbons §4.3.C) — background for Lecture 8.
- **Out** (`CLAUDE.md`): moral hazard — agency theory and incentive contracts with hidden effort. Efficiency wages and relational contracts appear **only** as repeated-game applications; M&R's comparison with incentive pay is one sentence.
- **Exercises are cited by number and page**, never reproduced.

Course-wide position

Lecture	Relation to this one
L1	the Cournot stage game and dominance — the one-shot problem collusion solves
L2	Nash equilibrium of Cournot and Bertrand: the punishments v^p
L3	backward induction: why the trust game fails once, and why a known end unravels
L4	multistage games and the one-stage deviation principle (Tadelis §9.5)
L5	finite vs infinite repetition, grim trigger, the folk theorem — **the toolkit used here**
L6	**collusion, reputation, efficiency wages: relational contracts**
L7	incomplete information: reputation becomes a belief about types
L8	signaling — the other way to make a promise credible

Bibliography actually used

Book	Where
Tadelis (2013) §10.2–10.5, pp. 192–209; exercises pp. 215–219	notes 01, 02, 05, 06
Gibbons (1992) §2.3.C–D, pp. 102–112	notes 02, 04, 07
Cabral (2017) Ch. 9, pp. 217–231; exercises pp. 245–246	notes 03, 04
Milgrom & Roberts (1992) Ch. 8, pp. 248–268; exercises pp. 285–286	notes 05, 06, 07
MWG §12.D, pp. 400–405	note 03 (G)
Muñoz-Garcia & Toro-Gonzalez (2019) Ch. 5 Ex. 4–5, 8 (pp. 228–239); Ch. 6 Ex. 2–5 (pp. 278–290), Ex. 10 (p. 310)	drill, cited in each note

Related project material. Rules: rules/06 (the "Relational and implicit contracts" section is still a placeholder), rules/game-theory. Reputation as a substitute for commitment (M&R pp. 139–140) in the Aula 2 notes, `Aula/notes-aulas-01-02/part2.tex`; efficiency wages named in `Aula/aula-01-notes.md` (Ford). No lists, quizzes or videos for Lecture 6 yet. Study guide: Map/study-guide (Lecture 6 block).

10.2–10.3 recap · One inequality runs the whole lecture

> **Claim.** Everything in Lecture 6 — a cartel, a trusted merchant, a well-paid worker — is the same check: **the one-period gain from cheating must not exceed the discounted loss of the relationship it destroys.** Written once, as Tadelis's (10.5), it is $\delta \geq (v^d - v^c)/(v^d - v^p)$; every later note only fills in v^c, v^d and v^p. Prev: — · Next: L06-02-cournot-collusion · Map: L06-00-map

> **What this note is not.** Finite versus infinite repetition, the folk theorem and the proofs belong to Lecture 5. Only what Lecture 6 uses is restated.

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1 · U — why the future can discipline the present

A stage game with a unique Nash equilibrium repeated a **known, finite** number of times unravels: in the last period only the stage Nash equilibrium can be played, so nothing that happens earlier can be rewarded or punished, and backward induction repeats the argument to period 1 (Tadelis Prop. 10.1, p. 192). Infinite repetition — or, equivalently, a probability δ that the game goes on after every period — removes the last period. Then the **proposed equilibrium itself** can serve as the reward ("carrot"), and the repeated stage Nash equilibrium as the punishment ("stick"): Tadelis calls it a *bootstrap equilibrium* (p. 197).

> **Two things are needed and both are examinable** (Tadelis p. 204): **no predetermined terminal period**, and **a discount factor that is not too small.** Drop either and cooperation that is not a stage-game best response cannot be sustained.

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2 · C — the tools, as Tadelis states them

Object	Statement	Page
Present value (Def. 10.2)	$v_i = \sum_{t=1}^{\infty} \delta^{t-1} v_i^t$	p. 193
Average payoff (Def. 10.3)	$\bar{v}_i = (1-\delta) \sum_{t=1}^{\infty} \delta^{t-1} v_i^t$ — a constant stream v has average v	p. 194
Prop. 10.2	playing a stage Nash equilibrium after **every** history is an SPE for any $\delta < 1$	p. 196
Prop. 10.3 (one-stage deviation)	a profile is an SPE **iff** no player gains by deviating at **one** history and conforming afterwards	p. 198
Grim trigger	play the cooperative action while every past outcome was cooperative; after any deviation play the stage Nash equilibrium forever	p. 198

Why Prop. 10.3 makes the check finite. Under grim trigger every history falls in one of two classes (p. 199): *no deviation yet* (play cooperatively) or *some deviation* (play stage Nash). Off the path the prescription is a stage Nash equilibrium followed by more of the same, so a one-shot deviation only lowers today's payoff — **no condition**. On the path there is one condition, the inequality of §3.

3 · C — the template, every step

Let v_i^c be player i's per-period payoff on the cooperative path, v_i^d the best payoff he can grab in one period by deviating, and v_i^p his per-period payoff in the punishment (the stage Nash equilibrium, v_i^* in Tadelis). Conforming forever is worth

$$v_i^c + \delta v_i^c + \delta^2 v_i^c + \cdots = v_i^c + \frac{\delta v_i^c}{1-\delta}.$$

Deviating today and being punished forever after is worth

$$v_i^d + \delta v_i^p + \delta^2 v_i^p + \cdots = v_i^d + \frac{\delta v_i^p}{1-\delta}.$$

No profitable one-shot deviation requires

$$v_i^c + \frac{\delta v_i^c}{1-\delta} \;\geq\; v_i^d + \frac{\delta v_i^p}{1-\delta}.$$

Multiply by $1-\delta > 0$:

$$(1-\delta)v_i^c + \delta v_i^c \;\geq\; (1-\delta)v_i^d + \delta v_i^p \iff v_i^c \;\geq\; (1-\delta)v_i^d + \delta v_i^p.$$

Collect δ: $v_i^c - v_i^d \geq -\delta v_i^d + \delta v_i^p = -\delta(v_i^d - v_i^p)$, and divide by $v_i^d - v_i^p > 0$:

$$\boxed{\delta \;\geq\; \delta_i^* \equiv \frac{v_i^d - v_i^c}{v_i^d - v_i^p}} \qquad \text{(Tadelis (10.5), p. 202)}$$

Read the fraction as temptation over stake. The numerator is the **one-period gain from cheating**; the denominator is the gain from cheating **plus** the per-period loss the punishment inflicts ($v^c - v^p$). Rearranged once more:

$$\underbrace{v^d - v^c}_{\text{gain today}} \;\leq\; \frac{\delta}{1-\delta} \underbrace{(v^c - v^p)}_{\text{loss per period, from tomorrow}}.$$

This second form is the one Milgrom & Roberts and the efficiency-wage model use (L06-05-reputation-trust, L06-07-efficiency-wages).

The middle line, $v^c \geq (1-\delta)v^d + \delta v^p$, is the **average-payoff form**: the cooperative payoff must beat a weighted average of the deviation and punishment payoffs. Every "conform vs deviate" figure in this unit plots exactly those two sides against δ.

3.1 · Worked — Tadelis's repeated Prisoner's Dilemma (pp. 197–200)

Stage game: (M, m) pays $(4, 4)$; F against m pays 5 to the finker and -1 to the other; (F, f) pays $(1, 1)$. So $v^c = 4$, $v^d = 5$, $v^p = 1$:

$$4 + \frac{4\delta}{1-\delta} \;\geq\; 5 + \frac{\delta}{1-\delta} \quad (10.3) \iff \delta \geq \frac{5-4}{5-1} = \frac{1}{4}.$$

δ	conform: $4/(1-\delta)$	deviate: $5+\delta/(1-\delta)$	verdict
0.20	5.000	5.250	deviate
0.25	**5.333**	**5.333**	**indifferent — the cutoff**
0.50	8.000	6.000	conform

---

4 · C — what δ stands for

δ is not only impatience. Tadelis reads it as the **probability that the game is played again** (p. 200); Cabral turns it into a formula with the interest rate, the **frequency** of price changes, the **hazard** that the industry ends and its **growth** — eq. (9.5), worked in L06-03-bertrand-market-structure §3. Everything that raises δ makes every relational contract in this lecture easier to sustain.

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5 · C — the checklist for any application

1. **Name the cooperative path** and its per-period payoff v^c for **each** player (asymmetric splits give different cutoffs — L06-02-cournot-collusion §2).
2. **Find the best one-period deviation** v^d: the stage best response to the others' cooperative actions (not "any" deviation — the most tempting one).
3. **Name the punishment** and its value v^p; check that it is itself an equilibrium (stage Nash forever is, by Prop. 10.2) — otherwise the threat is not credible.
4. **Apply (10.5)** for every player; the cooperative path survives iff $\delta \geq \max_i \delta_i^*$.
5. **Check off-path behaviour** with Prop. 10.3 (automatic under Nash reversion).

Links

- Next: L06-02-cournot-collusion (the first application). Map: L06-00-map.
- The basics in full: Lecture 5; subgame perfection and the one-stage deviation principle for finite games: Lecture 3, Lecture 4 (Tadelis §9.5).
- Rules: rules/game-theory item 4 ("repeated games are not one-shot games").
- Code: L06-11-python §1. Formal side: MWG §12.D (pp. 400–405), named in L06-03-bertrand-market-structure.
- Exercises: Tadelis 10.2 and 10.3 (p. 215) drill the cutoff; Muñoz-Garcia Ch. 6 Ex. 2 (from p. 278).

10.4 · Tacit collusion in quantities: who needs patience, and how much

Claim. A Cournot cartel is a relational contract between firms that cannot sign one: each firm produces less than its static best response **only because** cheating would trigger a return to Cournot forever. With linear demand and two firms the monopoly split survives iff $\delta \geq 9/17$; below that the cartel does not collapse to Cournot but **shrinks** to the most collusive output the threat can still support; and in an asymmetric split the firm with the **smaller** share needs the **more** patience. Prev: L06-01-recap-grim-trigger · Next: L06-03-bertrand-market-structure · Map: L06-00-map

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1 · U — collusion without a contract

Explicit cartel contracts are illegal (Sherman Act §1, 1890: "conspiracies in restraint of trade", Tadelis p. 201) and unenforceable anyway. The repeated game shows that firms may not need one: **shared beliefs about reward and punishment** support output restriction as a subgame-perfect equilibrium, with no meeting and no agreement — *tacit* collusion. Friedman (1971) was the first to show it, for exactly this Cournot case (Gibbons p. 102).

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2 · Worked — Tadelis's asymmetric split (pp. 201–202)

Stage game. Demand $p = 100 - q$, costs $c_i(q_i) = 10q_i$. From Tadelis §5.2.3:

Benchmark	Quantities	Price	Profit per firm
Cournot–Nash	$q_1^* = q_2^* = 30$	40	$v^* = 900$
Monopoly, split equally	22.5 each (total 45)	55	$(55-10)\times 22.5 = 1012.5$

The split Tadelis uses — deliberately unequal: $q_1^c = 22$, $q_2^c = 23$ (total 45, price 55):

$$v_1^c = (55-10)\times 22 = 990, \qquad v_2^c = (55-10)\times 23 = 1035.$$

Strategies (grim trigger). Each firm plays q_i^c in period 1 and keeps doing so while every past pair was $(22, 23)$; after any other history it plays $q_i^* = 30$ forever.

Off the path each firm expects 30 from the rival, and 30 is a best response to 30 — no condition, for any δ (p. 202).

On the path. The most tempting deviation is the static best response to the rival's cooperative output. From $\max_{q_i}(100 - q_i - q_j - 10)q_i$, the first-order condition $90 - 2q_i - q_j = 0$ gives

$$BR_i(q_j) = \frac{90 - q_j}{2} \qquad (10.4)$$

	Firm 1 (rival plays 23)	Firm 2 (rival plays 22)
deviation $q_i^d = BR_i(q_j^c)$	$(90-23)/2 = 33.5$	$(90-22)/2 = 34$
price	$100 - 33.5 - 23 = 43.5$	$100 - 22 - 34 = 44$
deviation profit v_i^d	$(43.5-10)\times 33.5 = \mathbf{1122.25}$	$(44-10)\times 34 = \mathbf{1156}$
conform v_i^c	990	1035
cutoff $\delta_i^* = \dfrac{v_i^d - v_i^c}{v_i^d - v^*}$	$\dfrac{132.25}{222.25} = \mathbf{0.595}$	$\dfrac{121}{256} = \mathbf{0.473}$

> **Erratum — check before you copy the book.** Tadelis prints firm 2's deviation profit as $(100-23-34)\times 34 - 10\times 34 = 1122$ and the cutoff $\delta_2 \geq 0.392$ (p. 202). The formula plugs in $q = 23$ — firm 2's **own** cooperative output — where the rival's output $q_1^c = 22$ belongs. With $q_1^c = 22$ the price is 44, $v_2^d = 1156$ and $\delta_2^* = 121/256 = 0.473$. **The book's conclusion survives**: $0.595 > 0.473$, so firm 1 is still the one that needs more patience. (Verified in L06-11-python §2.)

Why the smaller firm is the binding one (Tadelis p. 203, in his words "nicely reflects the temptations"): firm 1 **gains more** from deviating ($1122.25 - 990 = 132.25$ against $1156 - 1035 = 121$) **and loses less** when Cournot returns ($990 - 900 = 90$ per period against $1035 - 900 = 135$). Both effects push its cutoff up. The cartel survives iff $\delta \geq \max(0.595, 0.473) = 0.595$.

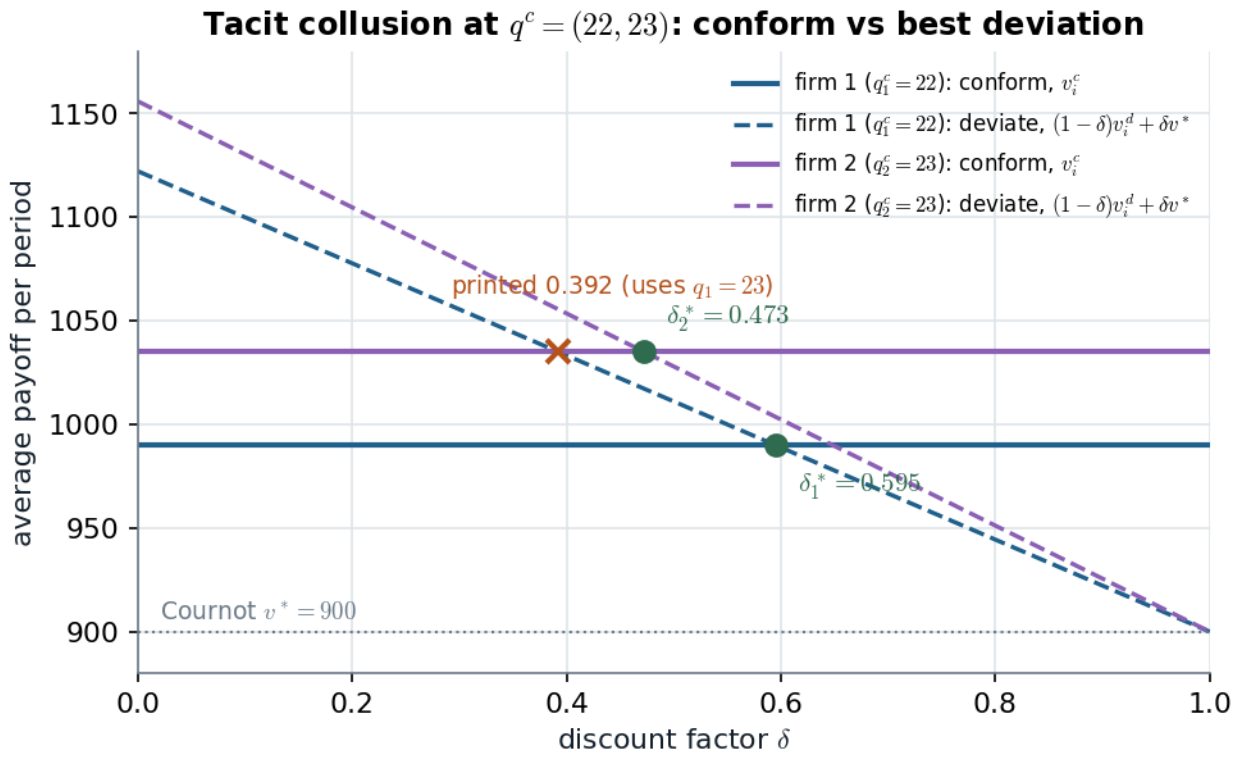


Figure — the average-payoff form of (10.5): each firm conforms where its flat line lies above its falling deviation line. Source: Tadelis §10.4, pp. 201–202, with v_2^d corrected to 1156.

Each dashed line starts at the deviation profit ($\delta = 0$: only today counts) and ends at Cournot's 900 ($\delta \to 1$: only the punishment counts). Firm 1's dashed line crosses its lower conform line later, so firm 1 sets the cartel's patience requirement. The cross shows where the printed 0.392 would put firm 2.

Lesson for designing a split. An unequal division of the monopoly profit raises the

cartel's required δ to the **worse-treated** firm's cutoff. A cartel that wants to survive at the lowest δ gives every member a share that equalizes the cutoffs, not the quantities (Cabral's asymmetry point, L06-03-bertrand-market-structure §4).

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3 · C — Gibbons's symmetric cartel: $\delta \geq 9/17$ (pp. 102–103)

Stage game. $P(Q) = a - Q$, marginal cost c, two firms. Cournot output $q_c = (a-c)/3$; monopoly output $q_m = (a-c)/2$; the cartel plays $q_m/2 = (a-c)/4$ each.

Payoff	Derivation	Value
collusive, $\pi_m/2$	price $a-(a-c)/2 = (a+c)/2$; margin $(a-c)/2$; times $(a-c)/4$	$(a-c)^2/8$
Cournot, π_c	margin $a - 2(a-c)/3 - c = (a-c)/3$; times $(a-c)/3$	$(a-c)^2/9$
deviation, π_d	$\max_{q_j}(a - q_j - \frac{a-c}{4} - c)q_j$: FOC $\frac{3}{4}(a-c) - 2q_j = 0$, $q_j = \frac{3}{8}(a-c)$; margin $\frac{3}{8}(a-c)$	$9(a-c)^2/64$

Grim trigger is an SPE iff (Gibbons (2.3.2), p. 103)

$$\frac{1}{1-\delta} \cdot \frac{\pi_m}{2} \;\geq\; \pi_d + \frac{\delta}{1-\delta}\,\pi_c.$$

Using the template, with $(a-c)^2$ cancelling:

$$\delta \;\geq\; \frac{\pi_d - \pi_m/2}{\pi_d - \pi_c} = \frac{\frac{9}{64} - \frac{8}{64}}{\frac{9}{64} - \frac{1}{9}} = \frac{\frac{1}{64}}{\frac{81-64}{576}} = \frac{1}{64} \cdot \frac{576}{17} = \boxed{\frac{9}{17} \approx 0.529}$$

Tadelis's numbers in Gibbons's model ($a = 100$, $c = 10$, $a-c = 90$): collusive 1012.5, deviation $q_j = 33.75$ with profit $33.75^2 = 1139.06$, Cournot 900 — $(1139.06-1012.5)/(1139.06-900) = 126.56/239.06 = 0.529$.

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4 · C — below 9/17 the cartel shrinks instead of dying (Gibbons pp. 103–104)

For $\delta < 9/17$ half the monopoly output cannot be sustained, but **repeating Cournot is always an SPE**, so the best grim-trigger cartel produces some $q^* \in [q_m/2, q_c]$. Take the trigger strategy "produce q^* while both always did; otherwise q_c forever".

- Conform: $\pi^* = (a - 2q^* - c)q^*$.
- Best deviation against q^*: $q_j = (a - q^* - c)/2$, profit $\pi_d = (a - q^* - c)^2/4$.
- Punishment: $\pi_c = (a-c)^2/9$.

The condition $\pi^* \geq (1-\delta)\pi_d + \delta\pi_c$ in units of $a-c$ (write q for $q^*/(a-c)$):

$$(1-2q)q \;\geq\; (1-\delta)\frac{(1-q)^2}{4} + \frac{\delta}{9}.$$

Multiply by 36: $36q - 72q^2 \geq 9(1-\delta)(1-2q+q^2) + 4\delta$. Move everything right:

$$\begin{aligned} 0 \;&\geq\; (72 + 9(1-\delta))q^2 - (36 + 18(1-\delta))q + 9(1-\delta) + 4\delta \\ \iff\; & 9(9-\delta)\,q^2 - 18(3-\delta)\,q + (9-5\delta) \;\leq\; 0. \end{aligned}$$

The quadratic opens upward, so the condition holds **between its roots**. One root is the Cournot output $q = 1/3$ (check: $9 - \delta - 6(3-\delta) + 9 - 5\delta = 0$). The product of the roots is $(9-5\delta)/(9(9-\delta))$, so the other root is that product divided by $1/3$:

$$\boxed{q^* = \frac{9-5\delta}{3(9-\delta)}\,(a-c)} \qquad \text{(Gibbons p. 104)}$$

It equals $(a-c)/3$ at $\delta = 0$ and $(a-c)/4$ at $\delta = 9/17$ ($\frac{9-45/17}{3(9-9/17)} = \frac{108/17}{432/17} = \frac{1}{4}$), and falls monotonically between.

Worked at $\delta = 1/2$, Tadelis's numbers. $q^* = \dfrac{9-2.5}{3\times 8.5} \times 90 = 0.2549 \times 90 = 22.94$ per firm; total 45.88, price 54.12, profit $(54.12 - 10) \times 22.94 = 1012.1$ — within half a unit of the monopoly share 1012.5, though the full share itself is not sustainable.

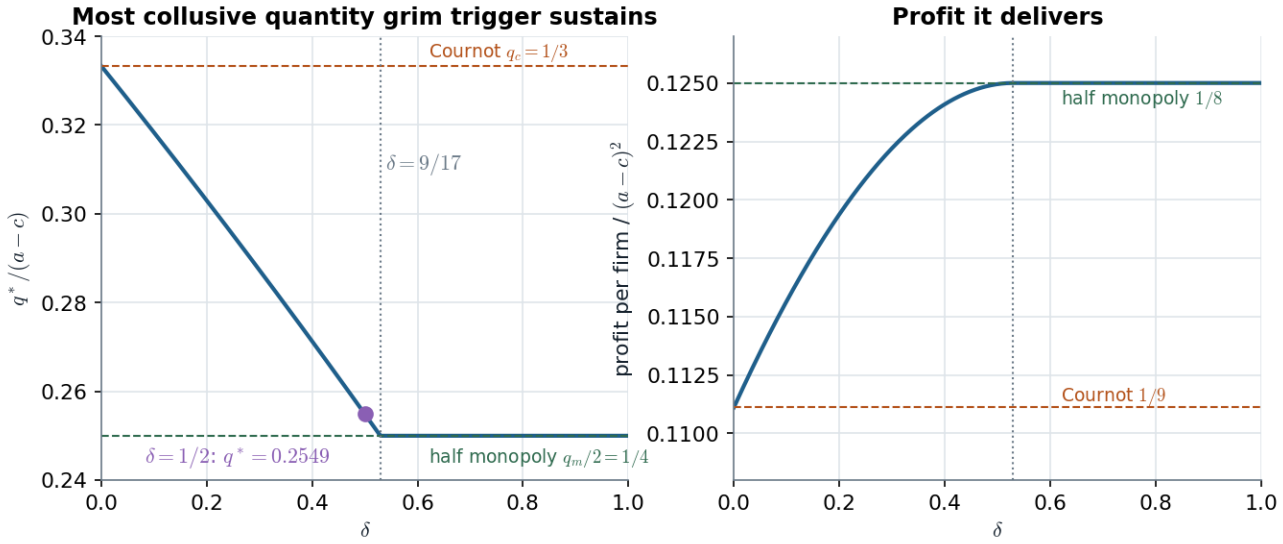


Figure — the most collusive symmetric output grim trigger sustains, and the profit it delivers. Source: Gibbons §2.3.C, p. 104, $q^* = (9-5\delta)/(3(9-\delta))\,(a-c)$.

Profit climbs steeply at low δ and flattens near 9/17: the last steps toward the monopoly output are worth little, because profit is flat at its maximum. That is why a cartel just short of the threshold loses almost nothing — and why the **carrot-and-stick** punishment of L06-04-punishments-price-wars §1, which reaches the full monopoly share at $\delta = 1/2$, gains little in profit but a lot in theory.

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5 · C — n Cournot firms

With n identical firms, $P = a - Q$, marginal cost c (the general case of Muñoz-Garcia Ch. 6 Ex. 4, p. 287):

Payoff per firm	Derivation	Value
collusive share	monopoly profit $(a-c)^2/4$ split n ways	$(a-c)^2/(4n)$
best deviation	rivals produce $(n-1)\frac{a-c}{2n}$; $q_d = \frac{1}{2}(a - c - \frac{(n-1)(a-c)}{2n}) = \frac{(n+1)(a-c)}{4n}$; margin equals q_d	$\frac{(n+1)^2(a-c)^2}{16n^2}$
Cournot	$q_N = \frac{a-c}{n+1}$, margin q_N	$(a-c)^2/(n+1)^2$

Numerator of (10.5): $\frac{(n+1)^2-4n}{16n^2}(a-c)^2 = \frac{(n-1)^2}{16n^2}(a-c)^2$. Denominator: $\frac{(n+1)^4-16n^2}{16n^2(n+1)^2}(a-c)^2 = \frac{\big((n+1)^2-4n\big)\big((n+1)^2+4n\big)}{16n^2(n+1)^2}(a-c)^2 = \frac{(n-1)^2(n^2+6n+1)}{16n^2(n+1)^2}(a-c)^2$. The ratio:

$$\boxed{\delta^*_{\text{Cournot}}(n) = \frac{(n+1)^2}{n^2+6n+1}} \qquad n=2:\ \tfrac{9}{17}=0.529, \quad n=3:\ \tfrac{16}{28}=0.571, \quad n=10:\ \tfrac{121}{161}=0.752.$$

It rises with n (more firms, smaller share, bigger relative temptation) and tends to 1. The side-by-side with Bertrand's $1 - 1/n$ is in L06-03-bertrand-market-structure §2.

> **Interactive companion.** The Cartel Patience Dial (`companions/companion-collusion-threshold.html`) — switch to Cournot, move n and δ, and watch the cutoff, the deviation profit and the most collusive sustainable output $q^*(\delta)$ respond; "snap to δ^*" lands exactly on the cutoff.

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6 · C — Tadelis's two closing remarks (p. 203)

1. **Beliefs without talk.** If firms share these beliefs without ever discussing them, they evade antitrust rules; where collusion is legal (OPEC) the same reward-and-punishment strategies replace an enforceable contract.
2. **Price wars are a puzzle for this model.** On the equilibrium path nobody deviates, so a price (quantity) war is **off-path** behaviour and should never be observed. The resolution — imperfect monitoring — is L06-04-punishments-price-wars §3.

A second misprint, harmless: Tadelis writes that the collusive price "will remain fixed (at $p = 45$ in the example previously solved)" (p. 203). Total output is 45; the price is $100 - 45 = 55$.

Links

- Template: L06-01-recap-grim-trigger §3. Next: L06-03-bertrand-market-structure.
- Static Cournot (the stage game): Lecture 2; its IESDS solution: Lecture 1.
- Code: L06-11-python §2–§3. Numbers: L06-09-formulas §2.1–2.2.

- Rules: rules/game-theory items 3–4 (multiplicity; patience).
- Exercises: Muñoz-Garcia Ch. 5 Ex. 4 (p. 228), Ex. 8 (p. 239); Ch. 6 Ex. 2 (p. 278), Ex. 4 (p. 287); Cabral Exercise 9.17 (p. 245).

Cabral 9.1, 9.3 · Price collusion: $\delta \geq 1 - 1/n$, and what moves δ

Claim. With prices, the stage game's Bertrand trap makes the punishment as harsh as it can be (zero profit) **and** the temptation as large as it can be (the whole market), so the cartel holds iff $\delta \geq 1 - 1/n$ — an all-or-nothing condition that does not depend on the price colluded on. Everything that raises the effective discount factor — **frequent interaction, a stable and growing market, few and similar firms, contact in many markets** — makes collusion easier; that list is the antitrust checklist. Prev: L06-02-cournot-collusion · Next: L06-04-punishments-price-wars · Map: L06-00-map

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1 · C — the repeated Bertrand duopoly (Cabral pp. 218–219)

Stage game. Homogeneous product, constant marginal cost c, no capacity constraints, simultaneous prices. The unique stage Nash equilibrium is $p = c$ with zero profit — the *Bertrand trap* (Cabral §8.1).

Grim strategies (Cabral p. 218; MWG calls them *Nash reversion*, p. 401). Set the monopoly price p_M in period 1; keep setting p_M while every past price was p_M; after any deviation, price at c forever.

Conform. Each firm gets half the monopoly profit π_M every period:

$$V = \tfrac{1}{2}\pi_M + \delta\,\tfrac{1}{2}\pi_M + \delta^2\,\tfrac{1}{2}\pi_M + \cdots = \frac{\frac{1}{2}\pi_M}{1-\delta} \qquad (9.1)\text{–}(9.2)$$

Deviate. Future payoff is zero whatever the deviation, so the best deviation maximizes today's profit: undercut to $p_M - \varepsilon$ and take the whole market,

$$V' = \pi_M \qquad (9.3)$$

(strictly π_M minus an arbitrarily small amount; Cabral's footnote c sets it equal).

Condition $V \geq V'$:

$$\frac{\frac{1}{2}\pi_M}{1-\delta} \geq \pi_M \iff \tfrac{1}{2} \geq 1-\delta \iff \boxed{\delta \geq \tfrac{1}{2}} \qquad (9.4)$$

In the template's terms, $v^c = \pi_M/2$, $v^d = \pi_M$, $v^p = 0$: $(\pi_M - \pi_M/2)/(\pi_M - 0) = 1/2$.

MWG's statement (C-level content, G-level proof). **Proposition 12.D.1** (p. 401): the Nash reversion strategies are an SPE of the infinitely repeated Bertrand duopoly iff $\delta \geq \frac{1}{2}$. **Proposition 12.D.2** (p. 403): for $\delta \geq \frac{1}{2}$ **any** price in $[c, p_M]$ can be supported the same way; for $\delta < \frac{1}{2}$ **every** SPE has all sales at $p = c$ in every period.

> **All or nothing.** Replace p_M by any $p \in (c, p_M]$: conform gives $\frac{1}{2}\pi(p)/(1-\delta)$, deviate gives $\pi(p)$, and $\pi(p)$ cancels. The cutoff is $\frac{1}{2}$ **whatever the price**. Contrast Cournot, where a shortfall in δ is absorbed by colluding *less* (L06-02-cournot-collusion §4). In Bertrand, a cartel short of patience has no fallback.

G — why nothing survives below $\frac{1}{2}$ (MWG p. 403, small type). In any SPE, at any date t each firm could undercut the lowest price, take (almost) the whole industry profit π_t today, and still guarantee itself at least zero afterwards. Conforming therefore requires $v_{jt} \geq \pi_t$ for $j = 1, 2$ (12.D.3), where v_{jt} is

firm j's equilibrium value from t on. Adding the two: $v_{1t} + v_{2t} \geq 2\pi_t$. At a date where industry profit is at its largest (case (i) of the proof), the two values together are at most $\pi_t + \delta\pi_t + \delta^2\pi_t + \cdots = \pi_t/(1-\delta)$. So $2\pi_t \leq \pi_t/(1-\delta)$, which fails for every $\pi_t > 0$ when $\delta < \frac{1}{2}$. (Case (ii), a supremum never attained, needs a limit argument.) Named, not examinable.

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2 · C — n firms: $\delta \geq 1 - 1/n$

With n symmetric firms the collusive share is π_M/n, the undercutter still takes π_M, and the punishment still yields zero:

$$\frac{\pi_M/n}{1-\delta} \geq \pi_M \iff \frac{1}{n} \geq 1-\delta \iff \boxed{\delta \geq 1 - \frac{1}{n}}$$

Cabral's reading (p. 227): "profit per firm would be smaller. However, the profit that a price cutter would gain would still be the same." MWG sets the J-firm version as Exercise 12.D.2 (p. 405).

n	Bertrand $1-1/n$	Cournot $(n+1)^2/(n^2+6n+1)$	Which needs more patience
2	**0.500**	**0.529**	Cournot
3	0.667	0.571	Bertrand
4	0.750	0.610	Bertrand
6	0.833	0.671	Bertrand
10	0.900	0.752	Bertrand

Why the ranking flips after $n = 2$. Bertrand punishes harder (zero, against Cournot's positive profit) — that lowers its cutoff; but it also tempts harder (the whole monopoly profit, against a best response that leaves the rivals some sales) — that raises it. At $n = 2$ the harsher stick wins by a hair; from $n = 3$ on, the bigger carrot for cheating dominates.

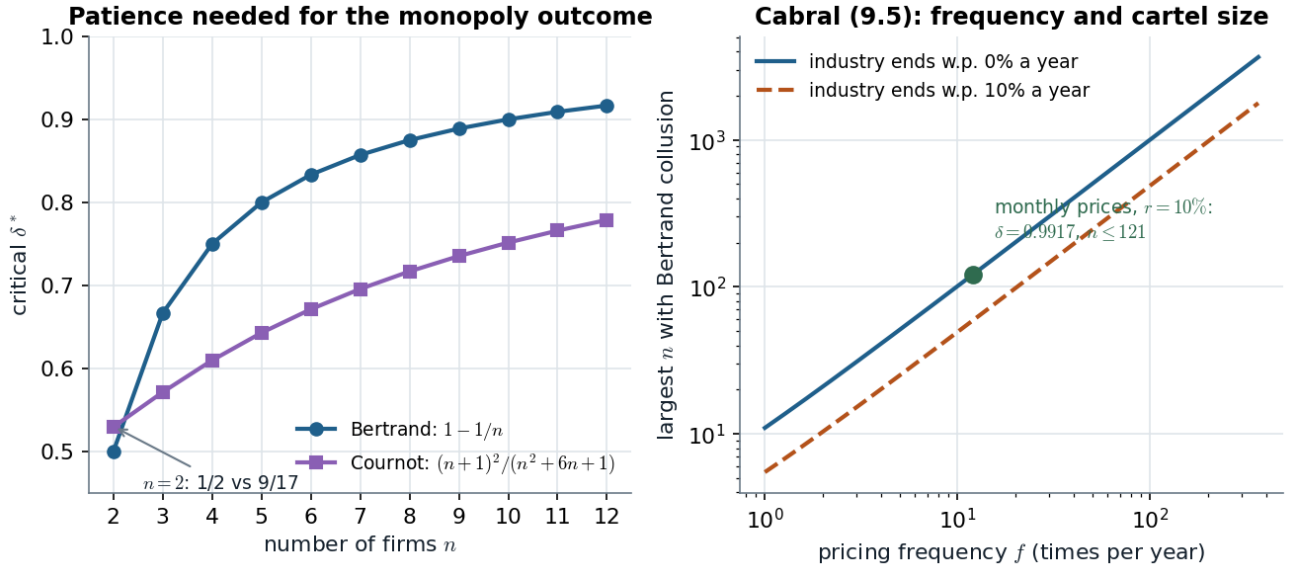


Figure — left, the two cutoffs by number of firms (Cabral p. 227; L06-02-cournot-collusion §5); right,

the largest Bertrand cartel $n \le 1/(1-\delta)$ when δ comes from Cabral (9.5) with $r = 10\%$ a year. Source: Cabral pp. 219–221, 227.

The right panel is the same inequality read backwards: $\delta \ge 1 - 1/n \iff n \le 1/(1-\delta)$. With $h = 0$ and $g = 0$, $1/(1-\delta) = 1 + f/r$, a straight line on log axes; a 10% annual chance that the industry disappears halves the sustainable cartel at every frequency.

> **Interactive companion.** The Cartel Patience Dial (`companions/companion-collusion-threshold.html`) — switch between Bertrand and Cournot at the same n and δ: the Bertrand cutoff jumps past Cournot's from $n = 3$ on, and below its cutoff the Bertrand cartel has no partial fallback.

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3 · C — what the discount factor is made of (Cabral pp. 219–220)

Cabral builds δ from four primitives:

Ingredient	Effect on the value of one unit of money next period	Factor
interest rate r per year, prices reset f times a year	one period is $1/f$ years: discount at r/f	$\dfrac{1}{1+r/f}$
hazard h: probability the industry is gone next period	the payoff arrives only with probability $1-h$	$(1-h)$
demand growth g per period	next period's profit is $(1+g)$ times bigger	$(1+g)$

$$\boxed{\delta = \frac{(1+g)(1-h)}{1+r/f}} \qquad (9.5)$$

Signs (p. 220): δ rises with f and g, falls with h and r. Hence Cabral's summary: **"Collusion is normally easier to maintain when firms interact frequently and when the probability of industry continuation and growth is high."** Gas stations that reprice daily collude more easily than holiday resorts that set annual rates; a pharmaceutical market with fast obsolescence (high h) less easily than cement (p. 220). MWG makes the same three points — growth, a probability of ending, and the **time it takes to detect a deviation** — on p. 403.

3.1 · Worked — "why don't firms collude more often?" (Cabral p. 221)

Annual interest 10%, monthly prices, $h = g = 0$:

$$\delta = \frac{1}{1+0.10/12} = \frac{1}{1.008\overline{3}} = 0.99174, \qquad n \le \frac{1}{1-\delta} = 1 + \frac{f}{r} = 1 + \frac{12}{0.10} = 121.$$

So "even if there are dozens of symmetric price setting firms", monopoly pricing is an equilibrium. With **annual** prices instead, $\delta = 1/1.1 = 0.909$ and $n \le 11$.

> **The model now over-predicts collusion** — the opposite of the static Bertrand trap. Cabral's four answers (p. 222): (1) **antitrust** makes explicit cartels illegal and can catch tacit

ones; (2) **turnover** — high entry and exit means a high h and a low δ; (3) the infinite price war is **not credible**: after a deviation both firms gain by renegotiating back to collusion, which removes the threat; (4) **prices are not perfectly observed**, so secret price cuts are possible. (3) and (4) are L06-04-punishments-price-wars §2–§3.

Box 9.1 (De Beers, p. 221) is the punishment made visible: when Zaire left the CSO in 1981, about a million carats of industrial diamonds flooded the market and the price fell from USD 3 to under USD 1.80 per carat; in 1983 Zaire asked to return, on worse terms.

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4 · C — structure that facilitates collusion (Cabral §9.3, pp. 227–231)

Factor	Mechanism in the model	Cabral's evidence
Few firms	$1 - 1/n$ rises with n: smaller share, same temptation	Exercise 9.17 (p. 245)
Symmetric firms	with a cost gap, the jointly efficient cartel gives the high-cost firm zero — it has **nothing to lose** and undercuts for any δ; an equal-price cartel instead tempts the low-cost firm, whose margin is large and whose punishment (rival pricing at its own higher cost) is weak	bromide cartel, six price wars 1885–1914 over **profit division** (p. 228)
Multimarket contact	identical markets: the stick and the carrot both double, cutoff unchanged; markets where each firm has its own cost advantage: the threat to retaliate **in the other market** sustains collusion a single market cannot	Box 9.3 airlines (p. 229); Box 9.4 dog food (p. 230)
Facilitating practices	most-favoured-customer clauses make a price cut to one buyer costly on all buyers — they lower v^d	Box 9.5 GE/Westinghouse price book, 1963–1975 (p. 231)

Multimarket contact, worked (p. 228–229). Two identical markets, each a Bertrand duopoly. Colluding in both: $2 \times \frac{1}{2}\pi_M/(1-\delta)$. Deviating in both (the best deviation, since any deviation triggers war everywhere): $2\pi_M$ today, zero after. Condition $\pi_M/(1-\delta) \geq 2\pi_M \iff \delta \geq \frac{1}{2}$ — identical to one market. Contact matters only when the markets differ.

Box 9.3, the number. On a route served by American, Delta and Northwest in 1988, the pairs met on 527, 357 and 323 of the top 1,000 routes: average contact $(527 + 357 + 323)/3 = 402.3$ — and higher contact predicts higher fares.

Links

- Previous: L06-02-cournot-collusion (quantities). Next: L06-04-punishments-price-wars.
- Static Bertrand: Lecture 2. Template: L06-01-recap-grim-trigger §3.
- Code: L06-11-python §4. Formulas: L06-09-formulas §1.2.
- Rules: rules/game-theory items 4 and 16 (outside options and participation — the "nothing to lose" firm).
- Exercises: Tadelis 10.8 (p. 218); Muñoz-Garcia Ch. 5 Ex. 5 (p. 231), Ch. 6 Ex. 5 (p. 290); Cabral 9.17 and 9.18 (p. 245); MWG Exercise 12.D.2 (p. 405).

Gibbons 2.3.C, Cabral 9.2 · Sharper sticks, and wars that happen on the path

Claim. Grim trigger is the simplest punishment, not the strongest: a **one-period** punishment harsher than Cournot, followed by a return to collusion, sustains the monopoly outcome at $\delta = 1/2 < 9/17$ and is easier to defend against renegotiation. And when firms see only noisy prices, **price wars are part of the equilibrium**, not evidence of cheating: no one deviates, yet bad luck must trigger punishment, or everyone would cheat. Prev: L06-03-bertrand-market-structure · Next: L06-05-reputation-trust · Map: L06-00-map

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1 · C — Abreu's carrot and stick (Gibbons pp. 104–106)

Two-phase strategy. Produce $q_m/2$ in period 1. In period t, produce $q_m/2$ if both firms produced $q_m/2$ in $t-1$, **or if both produced** x in $t-1$; otherwise produce x. So a deviation from either phase triggers **one** period of x (the *stick*); obeying the stick restores collusion (the *carrot*).

Normalize $a - c = 1$. Payoffs per firm:

Symbol	Meaning	Value
$\pi_m/2$	collusive	$1/8$
π_d	best deviation from collusion	$9/64$
$\pi(x)$	both produce x	$(1-2x)x$
$\pi_{dp}(x)$	best deviation from the punishment: $q_j = (1-x)/2$	$(1-x)^2/4$
$V(x)$	punishment now, collusion forever after	$\pi(x) + \dfrac{\delta}{1-\delta} \cdot \dfrac{\pi_m}{2}$

Collusive phase — prefer collusion forever to deviating and then being punished (2.3.3):

$$\frac{1}{1-\delta} \cdot \frac{\pi_m}{2} \;\geq\; \pi_d + \delta V(x).$$

Substitute $V(x)$: the right side is $\pi_d + \delta\pi(x) + \frac{\delta^2}{1-\delta}\frac{\pi_m}{2}$. Subtract $\frac{\delta^2}{1-\delta}\frac{\pi_m}{2}$ from both sides; the left becomes $\frac{1-\delta^2}{1-\delta}\frac{\pi_m}{2} = (1+\delta)\frac{\pi_m}{2}$:

$$(1+\delta)\frac{\pi_m}{2} \geq \pi_d + \delta\pi(x) \iff \boxed{\delta\Big(\frac{\pi_m}{2} - \pi(x)\Big) \;\geq\; \pi_d - \frac{\pi_m}{2}}$$

the gain from deviating today must not exceed the discounted loss from tomorrow's one-period punishment.

Punishment phase — prefer to administer the punishment than to deviate from it and face it again (2.3.4):

$$V(x) \;\geq\; \pi_{dp}(x) + \delta V(x).$$

With $V(x) = \pi(x) + \delta\frac{\pi_m/2}{1-\delta}$, multiply out $(1-\delta)V(x) \geq \pi_{dp}(x)$: $(1-\delta)\pi(x) + \delta\frac{\pi_m}{2} \geq \pi_{dp}(x)$, i.e.

$$\boxed{\delta\Big(\frac{\pi_m}{2} - \pi(x)\Big) \;\geq\; \pi_{dp}(x) - \pi(x)}$$

1.1 · Worked — $\delta = 1/2$ (Gibbons p. 106)

(2.3.3): $\frac{1}{2}(\frac{1}{8} - x + 2x^2) \geq \frac{9}{64} - \frac{8}{64} = \frac{1}{64}$. Multiply by 2: $2x^2 - x + \frac{1}{8} \geq \frac{1}{32}$, i.e. $2x^2 - x + \frac{3}{32} \geq 0$. Roots $x = \frac{1 \pm \sqrt{1-3/4}}{4} = \frac{1 \pm \frac{1}{2}}{4}$: $\frac{1}{8}$ and $\frac{3}{8}$. Holds for $x \leq \frac{1}{8}$ or $x \geq \frac{3}{8}$ — "$x/(a-c)$ is not between 1/8 and 3/8".

(2.3.4): left side $\frac{1}{16} - \frac{x}{2} + x^2$; right side $\frac{(1-x)^2}{4} - x + 2x^2 = \frac{1}{4} - \frac{3}{2}x + \frac{9}{4}x^2$. The inequality becomes $0 \geq \frac{3}{16} - x + \frac{5}{4}x^2$; roots $x = \frac{1 \pm \sqrt{1-15/16}}{5/2} = \frac{1 \pm \frac{1}{4}}{2.5}$: $\frac{3}{10}$ and $\frac{1}{2}$. Holds for $\frac{3}{10} \leq x \leq \frac{1}{2}$.

Both: $\boxed{\frac{3}{8} \leq \frac{x}{a-c} \leq \frac{1}{2}}$

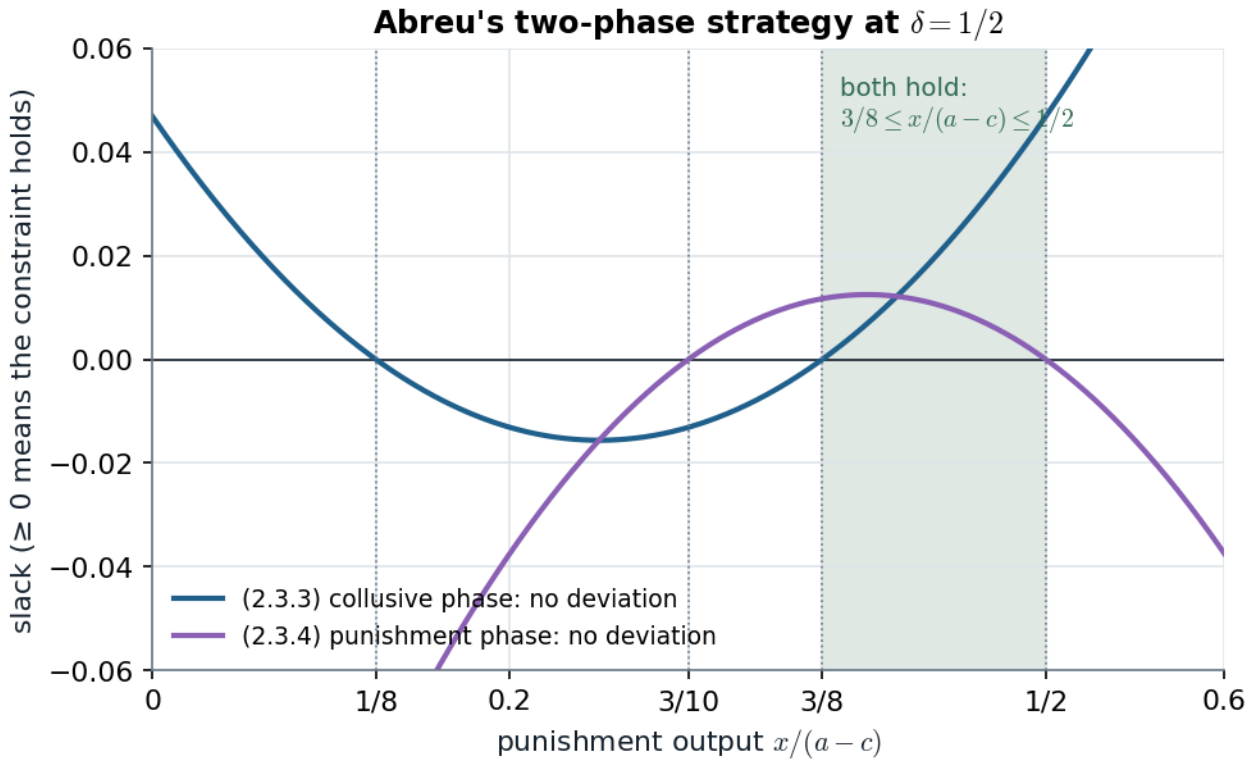


Figure — each curve is $\delta(\pi_m/2 - \pi(x))$ minus the right side of its constraint; the band is the set of punishments that work. Source: Gibbons pp. 105–106, $\delta = 1/2$.

Read the band. The stick must be **harsh enough** ($x \geq 3/8$, so the punishment profit $\pi(3/8) = \frac{1}{4} \cdot \frac{3}{8} = \frac{3}{32} = 0.094$ is below even Cournot's $1/9 = 0.111$) to deter the collusive-phase deviation, and **not so harsh** that the firms themselves refuse to carry it out ($x \leq 1/2$: at $x = 1/2$ total output is $a - c$, price equals c, profit zero). Since $1/2 < 9/17$, **the two-phase strategy reaches the monopoly outcome where grim trigger cannot** (L06-02-cournot-collusion §3).

> **Why it beats grim trigger.** Grim trigger's stick is Cournot **forever** — long but mild. Abreu's is a price war **for one period** — short but severe — and it is self-enforcing because obeying it is the ticket back to collusion. Severity per period, not duration, is what deters.

2 · C — renegotiation: is "forever" credible?

Gibbons (p. 111, for the efficiency-wage version) and Cabral (p. 222, his third reason) raise the same objection. After a deviation, grim trigger prescribes the stage Nash equilibrium forever. But **both** firms then prefer to return to collusion — so the deviator can propose "let bygones be bygones", and if both expect that, the punishment never happens and the deviation pays. The subgame-perfect equilibrium survives on paper; the **threat** does not survive negotiation.

Punishment	Hurts the punisher?	Renegotiation pressure
Grim trigger (stage Nash forever)	yes, forever	strong — both gain by restarting
Two-phase (one period of x, then collusion)	yes, for one period	weaker — collusion returns anyway after one period

G. The formal notion is renegotiation-proofness (introduced in Gibbons §2.3.A); it is named here, not examinable.

3 · C — imperfect monitoring: price wars on the equilibrium path

The puzzle (Tadelis p. 203). Under perfect monitoring a sustainable cartel never fights: punishment is off-path. Yet cartels have price wars.

The resolution (Tadelis pp. 203–204; Gibbons pp. 106–107; Cabral pp. 223–224). Firms do not see rivals' quantities or secret discounts; they see the **market price** (or their own sales), which also moves with demand. A low price is then a **noisy signal**: a rival cut, or demand fell. Each firm is tempted to cheat and plead *"I didn't deviate; demand must have been low!"* (Tadelis p. 203).

Cabral walks through the three candidate policies (p. 223):

Policy after low demand	Outcome
Never punish (assume it was demand)	not an equilibrium: each firm secretly cuts and blames the market
Infinite price war	deters cheating, but sooner or later a downturn triggers permanent war with nobody cheating
Price war for T periods, then back to collusion	with δ close enough to 1, some T deters cheating: **collusion and wars alternate, and no one ever cheats**

> **"If price cuts are difficult to observe, then occasional price wars may be necessary to discipline collusive agreements"** (Cabral p. 224). Wars are the price of incentive compatibility under noise — a "necessary evil" — not a breakdown.

Gibbons's version: Green and Porter (1984) *trigger-price* equilibria — any price below a critical level starts a Cournot phase; in equilibrium no firm deviates, yet an especially bad shock triggers punishment; because punishments happen by accident, **finite** punishments beat infinite ones, and Abreu–Pearce–Stacchetti (1986) show two-phase strategies can be optimal (p. 107).

Box 9.2 — the Joint Executive Committee, 1880–1886 (Cabral p. 224). A legal railroad cartel on eastbound Chicago–Atlantic freight. Several times it thought cheating had occurred, cut prices for a while and returned to the collusive rate; the wars were caused by **unpredictable disturbances**, not by entry or by the predictable lake-navigation season.

G. Repeated games with imperfect monitoring are "extremely technical and beyond the scope" of Tadelis (p. 204; Mailath and Samuelson 2006). Named, not examinable; the drill is Muñoz-Garcia Ch. 6 Ex. 10 (p. 310) and Cabral Exercise 9.19 (p. 245).

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4 · C — observable demand shocks: wars in booms (Rotemberg–Saloner)

If demand varies but **everyone sees** the current shock before pricing (Gibbons p. 106; Cabral p. 225), deviations are detectable, yet the temptation varies: the gain from undercutting is proportional to **current** demand, while the future loss (with i.i.d. shocks) is the same in every state. The no-deviation constraint binds in **high-demand** periods, so the cartel must **lower its price in booms** to keep members loyal.

	Green–Porter (unobserved shocks)	Rotemberg–Saloner (observed shocks)
when wars/low prices come	**after** low demand	**during** high demand
price over the cycle	pro-cyclical	**counter-cyclical**
Cabral's evidence	JEC railroads, 1880–1886	cement 1947–1981: +1% GDP, relative price −0.5% to −1% (p. 225)

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5 · U — a firm whose δ falls starts the war (Cabral pp. 225–227)

A firm in financial trouble attaches a higher probability to exiting — a higher hazard h in (9.5), so a lower δ. If its δ drops below its cutoff, it deviates and a price war follows: "Fares are dictated not by the strongest, but by the financially troubled" (airline executive, p. 226). Wars can also be started by the **strong**: the *Times* cut its price from 45 to 30 pence in July 1993 and to 20 pence ten months later; Google cut cloud prices by 30–85% in March 2014 (pp. 226–227).

Links

- Previous: L06-03-bertrand-market-structure. Next: L06-05-reputation-trust.
- Template and one-stage deviation: L06-01-recap-grim-trigger; renegotiation in the employment relation: L06-07-efficiency-wages §5.
- Code: L06-11-python §5. Pitfalls: L06-10-pitfalls items 5–6.
- Exercises: Muñoz-Garcia Ch. 6 Ex. 3 (temporary punishments, p. 282), Ex. 10 (imperfect monitoring, p. 310); Cabral 9.19 (p. 245) and 9.20 (p. 246).

10.5.1, M&R pp. 259–266 · Reputation: trust that pays for itself

> **Claim.** In the one-shot trust game trust is never offered, because it would never be honoured. Repeat it and the trusted party honours trust **to keep being trusted**: honesty becomes an asset whose value is the stream of future business, and it is worth protecting iff $\delta \geq G/(V+G)$ — the temptation over the temptation plus the per-period value of the relationship. A *reputation* is that asset when the future partners are **different people who observe the past**. Its limits are the limits of the assumption that everyone can tell, cheaply and after the fact, whether the right thing was done. Prev: L06-04-punishments-price-wars · Next: L06-06-institutions-reputation · Map: L06-00-map

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1 · C — Tadelis's trust game (pp. 204–205)

One shot (Fig. 10.1). Player 1 trusts (T) or not (N); if trusted, player 2 cooperates (C) or defects (D). Payoffs (player 1, player 2): $N \to (0,0)$, $(T,C) \to (1,1)$, $(T,D) \to (-1,2)$. Backward induction: player 2 defects ($2 > 1$); anticipating it, player 1 does not trust ($0 > -1$). The SPE outcome $(0,0)$ is Pareto-dominated by $(1,1)$.

Infinitely repeated (footnote 8: the stage game is sequential, but every extensive form has a normal form, so the repeated-game machinery applies). Grim trigger:

- **Player 1:** trust in period 1, and keep trusting while no deviation from (T,C) has occurred; after one, never trust again.
- **Player 2:** cooperate in period 1, and keep cooperating while no deviation from (T,C) has occurred; after one, defect forever.

Off the path each prescription is a best response to the other's (never trust; defect if ever trusted). **On the path** player 1's stream of 1s beats 0s. Player 2 conforms iff the stream of 1s beats 2 today and 0 after:

$$\frac{1}{1-\delta} \;\geq\; 2 \iff 1 \geq 2-2\delta \iff \boxed{\delta \geq \tfrac{1}{2}} \qquad \text{(p. 205)}$$

Template: $v^c = 1$, $v^d = 2$, $v^p = 0$, so $\delta^* = (2-1)/(2-0) = \frac{1}{2}$.

> **Footnote 9, the trap.** Writing player 2's strategy as "cooperate as long as player 1 trusts me" gives a Nash equilibrium that is **not** subgame-perfect — some off-path prescriptions are not best responses. Tadelis leaves the proof as Exercise 10.4 (p. 216). The lesson: a repeated-game strategy must say what to do **after every history**, including the ones the equilibrium never reaches.

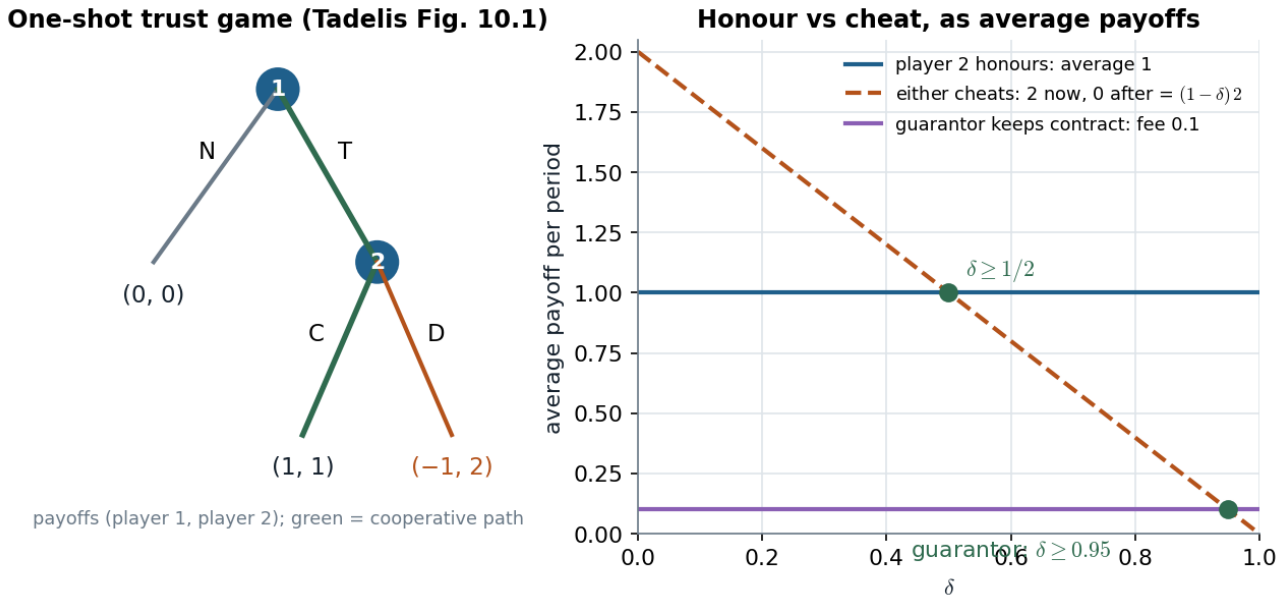


Figure — the stage game and its two repeated-game thresholds: $\delta \geq 1/2$ for the trusted player (p. 205) and $\delta \geq 0.95$ for a guarantor paid 0.1 a period who could pocket a bond of 2 (p. 207, L06-06-institutions-reputation §1). Source: Tadelis Fig. 10.1, §10.5.1–10.5.2.

The same dashed "cheat" line serves both players because both grab 2 once and get 0 after; what differs is the flat line they give up — the trusted player's 1 against the guarantor's 0.1. A thinner stream of rents needs a far more patient keeper.

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2 · C — Milgrom & Roberts's elementary theory (pp. 259–263)

Setting. Relational contracts — an authority relation, an employee who accepts direction "within somewhat nebulous customary limits", a contractor who must handle rain delays and discontinued fixtures (pp. 259–261) — leave one party relying on the other's good faith. The **elementary theory** assumes contracts cannot specify what to do, **but the parties can tell afterwards whether the right thing was done** (p. 261).

The box (p. 261): the offeror offers trust or not; the decision maker honours it or not.

Offeror Decision maker	Honour trust	Don't
Offer trust	$V,\ V$	$-L,\ V+G$
Don't offer	$0,\ 0$	$0,\ 0$

with $V, G, L > 0$: honoured trust beats no trade ($V > 0$), dishonouring pays G extra, and the offeror risks L. One shot: no trust (p. 262), the same unravelling as Tadelis's tree.

The decision maker's speech (p. 262): "trust me once … if I should ever fail to honor your trust, you would never trust me again. For that very reason, it will actually be in my interest … to honor your trust." Is it believable? Value a stream of X per meeting as NX today:

$$\text{honour forever: } NV, \qquad \text{cheat now: } V+G \text{ today, nothing after.}$$

M&R's multiplier (footnote 11, p. 263): meetings every t periods at interest r,

$$N = \frac{1}{(1+r)^t - 1} + 1 = \frac{(1+r)^t}{(1+r)^t - 1} = \frac{1}{1-(1+r)^{-t}} = \frac{1}{1-\delta}, \qquad \delta \equiv (1+r)^{-t}.$$

So the exact condition is

$$NV \geq V + G \iff (N-1)V \geq G \iff \frac{\delta}{1-\delta} V \geq G \iff \boxed{\delta \geq \frac{G}{V+G}}$$

— Tadelis's $\frac{1}{2}$ when $V = G = 1$.

> **M&R write the comparison as** $NV > G$ (p. 263) — honouring is worth NV against "only G" from cheating. Their box gives the cheater $V + G$ in the round he cheats; the exact one-shot-deviation comparison is $NV \geq V + G$. The two differ by one round's V and agree in spirit (and almost in value when N is large). In an exam, derive from the box.

2.1 · Worked — how often must you meet? ($r = 10\%$ a year)

Meetings	$\delta = (1.1)^{-t}$	$N = 1/(1-\delta)$	honours iff $G \leq (N-1)V$
every year ($t = 1$)	0.909	11.0	$G \leq 10V$
every 5 years ($t = 5$)	$1/1.6105 = 0.621$	2.64	$G \leq 1.64V$
every 20 years ($t = 20$)	$1/6.727 = 0.149$	1.17	$G \leq 0.17V$

Frequency is patience: a relationship renewed every year carries a temptation ten times its per-meeting value; one renewed every twenty years almost none (compare Cabral's f in L06-03-bertrand-market-structure §3).

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3 · C — many equilibria (M&R p. 263)

The trusting equilibrium is not the only one. **Never offer, never honour** is also an equilibrium (it repeats the stage SPE). So is a third: the decision maker **cheats every second time** trust is offered and the offeror keeps offering as long as there are never two cheats in a row — an equilibrium if $L < V$ and rounds are frequent (the offeror's average over a pair of rounds is $(V - L)/2 > 0$). "The hypothesis of Nash equilibrium rules out some patterns of behavior but does not always lead to a unique prediction" (p. 263) — the folk theorem of Lecture 5 in M&R's words.

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4 · C — from relationship to reputation (M&R p. 263)

Nothing in §2 needs the offeror to be the **same** person each round. If the decision maker faces a **series of different offerors**, each of whom trusts iff trust was honoured the last time it was offered, his calculation is unchanged: he honours trust "to encourage future trading partners to offer trust or, in other words, to maintain his or her **reputation for honesty**." Two preconditions follow, and both are where reputations fail:

1. **Future partners observe the past** (the history must travel).
2. **They can judge it** — tell honoured from dishonoured trust.

A good reputation then attracts partners **and saves the cost of detailed contracts and of legal enforcement**.

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5 · U — "live and let live", 1914–1918 (M&R p. 264)

The same opposing battalions faced each other for long stretches on the Western Front: the repeated game. Each side could reciprocate deadly fire, so "fire in earnest only if fired upon" became an equilibrium — and to keep it one, each side **showed it could retaliate** (German snipers cutting a hole in a cottage wall; artillery placing a few accurate shots) while showing it would not fire unprovoked (Axelrod 1984). Commanders ended the truces with commando raids that set off cycles of retaliation. **The punishment must be visibly feasible**, and an outsider who can inject deviations can destroy the equilibrium.

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6 · C — the limits of reputation (M&R pp. 264–266)

Limit	Why it bites	What helps
Ambiguity — was it honourable?	if right behaviour were easy to judge, courts could enforce it cheaply; parties disagree about circumstances and about what would have worked	shared principles
Outsiders must judge	a reputation system asks a *series* of third parties to evaluate disputes they did not see — costly and error-prone	institutions that judge and broadcast (L06-06-institutions-reputation)
The end game	the relationship must be expected to last long enough that its value exceeds the gain from cheating; near a known end, reputation is worth little	explicit incentives, sale of the business

Corporate culture as a fix for ambiguity (p. 265): simple, communicable rules that make "the right thing" unambiguous — the 49ers' rule that an injured starter always gets his place back (so injuries are not concealed); Nordstrom's story of cheerfully accepting returned tyres it never sold. **Reputations and the law** (pp. 265–266): in Shasta County, neighbours settle stray-cattle damage by community norms, not the law (Ellickson); "most contractual disputes never reach a court of law."

The end game, three cases (p. 266):

- **CEOs near retirement**: career concerns weaken, so pay should become more sensitive to performance — Gibbons and Murphy found exactly that.
- **Selling the business**: a founder near retirement keeps the firm's reputation intact to get a higher price — the firm acts with a longer horizon than its owner (the brand-name logic of L06-06-institutions-reputation §2).

- **Bankruptcy**: customers of a failing firm claim the last deliveries were defective and refuse to pay in full — with the relationship ending, attitudes harden.

  The end game is the finite-horizon unravelling of Lecture 5 in business form: the final period cannot be rewarded, so it cannot be disciplined, and the problem creeps backwards. M&R Quantitative Exercise 4 (p. 286) asks it for a trust game played exactly twice.

Links

- Previous: L06-04-punishments-price-wars. Next: L06-06-institutions-reputation.
- Finite vs infinite repetition and the folk theorem: Lecture 5. Backward induction in the centipede game, the same unravelling: `Listas/pratica-aula-01-02-solutions/item19.tex`.
- Reputation as a substitute for commitment (M&R pp. 139–140), already in the Aula 2 notes: `Aula/notes-aulas-01-02/part2.tex`, section "Commitment, reputation and the ultimatum game".
- Rules: rules/06 ("Relational and implicit contracts"), rules/game-theory items 4 and 15.
- Code: L06-11-python §6. Synthesis: L06-08-relational-contracts.
- Exercises: Tadelis 10.4 (p. 216), 10.12 (p. 219); M&R Quantitative Exercises 4–5 (p. 286).

10.5.2–10.5.3, M&R pp. 266–268 · When the traders are strangers: borrowed reputations

> **Claim.** One-shot traders cannot discipline each other, but a **long-lived third party** can: a guarantor holding a bond, a judge, a brand name passed from owner to owner. The repeated game moves from the traders to the institution, and the institution's own no-cheating condition is the same inequality — with a thin per-period rent it needs a very high δ (0.95 in Tadelis's example). Reputation is then not a trait of a person but an **asset that can be held, sold and lost**. Prev: L06-05-reputation-trust · Next: L06-07-efficiency-wages · Map: L06-00-map

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1 · C — the guarantor (Tadelis pp. 205–207)

A bond fixes the one-shot game (Fig. 10.2). Player 2 deposits 2 with a guarantor (player 3) under the contract "if player 1 trusts me and I defect, you keep the 2; otherwise return it". If the guarantor **must** follow the contract, defecting nets player 2 $2-2=0<1$: he cooperates, player 1 trusts, and the unique SPE is cooperative trade.

Two objections (p. 206): why would such a committed guarantor exist; and if player 3 is a real player, his dominant strategy is to **keep** the deposit.

The repeated-game answer. Player 3 is a long-lived local official — a judge, an arbitrator — with discount factor δ. In each period t a **new** pair P_1^t, P_2^t arrives and plays the trust game once. P_2^t may post a bond of 2 with player 3 for a fee $w=0.1$.

Player	Strategy (Tadelis pp. 206–207)
P_1^t	trust iff P_2^t posted a bond
P_2^t	post a bond iff player 3 has honoured the bond contract in every earlier period; if a bond was posted, cooperate, otherwise defect
player 3	return the bond to a P_2^t who cooperated; keep it otherwise

Traders. Given player 3's strategy, each pair plays Fig. 10.2: with the bond, P_2^t gets $1-0.1=0.9$ by cooperating against $2-2-0.1=-0.1$ by defecting — he cooperates and posts the bond ($0.9>0$).

The guarantor. Honour: 0.1 every period. Cheat: keep the bond, 2 once, then no trader ever posts again:

$$\frac{0.1}{1-\delta} \;\geq\; 2 \iff 0.1 \geq 2-2\delta \iff \boxed{\delta \geq 0.95} \qquad \text{(p. 207)}$$

Template: $v^c=0.1$, $v^d=2$, $v^p=0$: $(2-0.1)/2=0.95$.

> **Footnote 10, the design margin.** The bond need only exceed **1**, player 2's one-period gain from defecting; and a trader would pay any fee up to **1** for the service. Fee and bond are the institution's two dials: a higher fee raises the guarantor's stake in his reputation, a smaller bond lowers his temptation. Exercise 10.10 (p. 219) pushes this to its limit.

History (p. 207): Milgrom, North and Weingast (1990) — the **law merchants** of medieval Europe as such third parties; future traders "somehow learned about the law merchant, and about his past adherence" to the norms. Greif (2006) surveys institutions and trade.

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2 · C — brand names: a reputation you can sell (Tadelis pp. 207–209)

Kreps (1990b). Drop the third party. Before period 1, P_2^1 can create a unique **brand name** ("Trusted Associates") and sell it to the next period's player 2 at a price $p^* > 1$. Strategies (p. 208), in outline: player 1 trusts a brand that was **bought** and under which **no abuse of trust ever occurred**; each player 2 buys the clean brand, cooperates, and sells it on at p^*.

Why it works (Tadelis: "The investment made in the name … acts like a bond"). The brand is worth p^* to its holder only while it is clean. Cheating earns one extra unit today (2 instead of 1) and destroys a resale value of $p^* > 1$. Short-lived owners become, through the name, an **infinitely lived** player — "the concept of a terminal period is removed altogether". Proving the profile is an SPE is Exercise 10.11 (p. 219).

Roots: Klein and Leffler (1981), a firm's brand value as a bond for its performance; Crémer (1986), a firm sustaining cooperation among employees who come and go (p. 208).

> **The open question** (pp. 208–209): bootstrap equilibria say nothing about **how** reputations arise or why their value rises and falls. Tadelis's remark: *The Princess Bride*'s Dread Pirate Roberts is a name handed from captain to captain — "no one would surrender to the Dread Pirate Westley."

M&R's product-market version (p. 259): Klein and Leffler — "thin profit margins are the enemy of quality"; with recognizable products and high margins, the value of staying in business outweighs the short-run gain from degrading quality. A brand covering a whole product line extends the reputation to products replaced too fast for their own track record to matter (the Black & Decker drill).

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3 · C — institutions that make reputations work (M&R pp. 266–268)

M&R's **advanced theory**: when people specialize, transactions with any one partner are infrequent and the elementary theory's repeated relationship is missing. Two responses: replace reputation with the **legal system** (general rules, slow, costly, sometimes without authority across borders), or **buttress** reputation with private institutions:

Institution	What it does in the model	M&R
Law merchant — private judges, *lex mercatoria*	**judges** disputes and **broadcasts** offenders: the outsider's two hard tasks (§6 of L06-05-reputation-trust)	p. 267
Whalers' "gam"	captains exchange information on disputes at sea; US whalers in a few towns needed none — no recorded court case over whale ownership until rents fell in the late 19th century	pp. 267–268

Institution	What it does in the model	M&R
Better Business Bureau, credit bureaus, consumer agencies	collect and **disseminate** past behaviour, raising the cost of cheating	p. 268
Merchant guilds, the Hansa	coordinate **sanctions** — an embargo of a city that reneged — against the free-rider problem of individual sanctioners	p. 268

The Hansa, worked as an incentive problem (p. 268). In 1280 the guild trading in Bruges moved its trade to Aardenburg; Bruges broke the embargo by offering special terms to merchants from various towns. Each merchant's gain from breaking the embargo was private, the cost (a weaker threat) shared — a free-rider problem. The Hansa added a **second-order sanction**: traders who broke a declared embargo could have their **home city** embargoed, which made complying individually rational again.

The Maghribi traders (M&R pp. 252, 258; Greif 1989). Eleventh-century merchants in Fustat ostracized an agent who cheated **any** of them. In the efficiency-wage formula of L06-07-efficiency-wages §6 this **lowers the cheater's outside option** $\bar{w}$ and **raises the multiplier** N (all merchants' business is lost, not one), and the web of correspondence **raises detection** p. Overseas agency became cheaper to motivate as the network grew.

Links

- Previous: L06-05-reputation-trust. Next: L06-07-efficiency-wages.
- Institutions and incomplete contracts: rules/06.
- Code: L06-11-python §6 (the 0.95 check). Formulas: L06-09-formulas §2.4.
- Exercises: Tadelis 10.10, 10.11 (p. 219).

Gibbons 2.3.D, M&R pp. 250–255 · Efficiency wages: a rent the worker will not risk

Claim. When effort cannot be contracted on, a firm can still buy it with a **promise**: pay more than the worker's outside option and fire him if output is ever low. The wage must leave a **rent** large enough that the expected loss of the job outweighs the one-period gain from shirking — $w^* = w_0 + e + \frac{(1-\delta)e}{\delta(1-p)}$. The premium explodes as detection fails ($p \to 1$) or patience vanishes ($\delta \to 0$), and the relationship survives only if the job produces enough to pay it. This is the cartel inequality again, with the worker as the party tempted to cheat. Prev: L06-06-institutions-reputation · Next: L06-08-relational-contracts · Map: L06-00-map

Scope. Covered as a **repeated-game application** only. Incentive contracts with hidden effort — agency theory, optimal pay-for-performance — are moral hazard, **out of scope** (`CLAUDE.md`); M&R's comparison with incentive pay (§7) is named, not developed.

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1 · U — why pay more than the market

M&R open with an organization that must rely on an employee's honesty — a police officer offered bribes, a tanker captain who must stay sober, an analyst who must investigate with diligence — and whose only sanction is **dismissal** (fines are impractical when wealth is limited, p. 250). Then the threat of firing is the incentive, and it bites only if the job is worth more than the next one: the worker must earn a **rent**. Marshall's epigraph says it: "Highly paid labour is generally efficient and therefore not dear labour" (p. 248). Lord Clive, reforming the East India Company's servants around 1765, saw it too: "it was absurd to give men power and to require them to live in penury" (p. 252). The Aula 1 notes meet the same logic in Ford's five-dollar day (`Aula/aula-01-notes.md`).

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2 · C — Gibbons's stage game (pp. 107–108)

1. The firm offers a wage w.
2. The worker accepts or rejects; rejecting means self-employment at w_0.
3. If he accepts, he supplies effort (disutility e) or shirks (no disutility). **Effort is not observed; output is.** With effort output is $y > 0$ for sure; shirking gives y with probability p and 0 with probability $1 - p$. Low output is therefore proof of shirking.

Payoffs when employed at w: firm $y - w$ (or $-w$ if output is 0), worker $w - e$ (or w if he shirks). **Assumptions** (p. 108):

$$y - e > w_0 > py$$

— employment with effort is efficient, and self-employment beats employment with shirking.

One shot. The wage is paid in advance, so the worker shirks; anticipating it, the firm offers $w \leq w_0$ and the worker stays self-employed. The efficient relationship never forms.

Why not pay on output? Courts may be unable to verify output — its quality, unexpected production difficulties — so output-contingent contracts are infeasible or imperfect, "but there will remain a role for the repeated-game incentives studied here" (p. 108).

3 · C — the worker's condition (Gibbons pp. 108–110)

Strategies. Call the history *high-wage, high-output* if every past offer was w^*, every offer was accepted and every output was high.

- **Firm:** offer w^* in period 1 and after every high-wage, high-output history; offer $w = 0$ otherwise.
- **Worker:** accept iff $w \geq w_0$; supply effort iff the history, **including the current offer**, is high-wage, high-output; shirk otherwise. (The stage game is sequential, so a firm deviation is detected *within* the stage and answered at once, p. 109.)

Effort forever. Output is high for sure, so the same choice recurs next period:

$$V_e = (w^* - e) + \delta V_e \;\Rightarrow\; V_e = \frac{w^* - e}{1-\delta}.$$

Shirking. This period he gets w^* (no effort cost). With probability p output is high and the same situation recurs; with probability $1-p$ he is fired and self-employed forever:

$$V_s = w^* + \delta\Big[p\,V_s + (1-p)\frac{w_0}{1-\delta}\Big] \;\Rightarrow\; V_s(1-\delta p) = w^* + \frac{\delta(1-p)w_0}{1-\delta} \;\Rightarrow\; V_s = \frac{(1-\delta)w^* + \delta(1-p)w_0}{(1-\delta p)(1-\delta)}.$$

Effort iff $V_e \geq V_s$. Multiply both by $(1-\delta)(1-\delta p) > 0$:

$$(w^* - e)(1-\delta p) \;\geq\; (1-\delta)w^* + \delta(1-p)w_0.$$

Collect w^*: $w^*\big[(1-\delta p) - (1-\delta)\big] = w^*\,\delta(1-p)$, so

$$w^*\,\delta(1-p) \;\geq\; e(1-\delta p) + \delta(1-p)w_0 \iff w^* \;\geq\; w_0 + \frac{1-\delta p}{\delta(1-p)}\,e.$$

Split $1-\delta p = \delta(1-p) + (1-\delta)$:

$$\boxed{w^* \;\geq\; w_0 + e + \frac{1-\delta}{\delta(1-p)}\,e} \qquad (2.3.5)$$

The firm must cover the **outside option** w_0, the **effort cost** e, and a **wage premium** $\frac{(1-\delta)e}{\delta(1-p)}$ — the rent.

Case	Premium	Reading
$p \to 1$ (shirking almost never caught)	$\to \infty$	"the wage premium must be extremely high" (p. 110)
$p = 0$ (perfect monitoring)	$\frac{1-\delta}{\delta}e$	(2.3.6): $w^* \geq w_0 + \big(1 + \frac{1-\delta}{\delta}\big)e = w_0 + e/\delta$
$\delta \to 1$	$\to 0$	a patient worker needs almost no rent

Case	Premium	Reading
$\delta \to 0$	$\to \infty$	no future, no discipline

The template, recovered. Rearranging (2.3.5):

$$\underbrace{e}_{\text{gain today}} \;\leq\; \underbrace{(1-p)}_{\text{detection}} \cdot \frac{\delta}{1-\delta} \cdot \underbrace{(w^* - e - w_0)}_{\text{rent per period}}.$$

Exactly L06-01-recap-grim-trigger §3's second form, with the loss weighted by the probability that cheating is caught.

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4 · C — the firm's condition and the patience required (p. 110)

Given the worker's strategy, the firm compares paying w^* forever ($y - w^*$ a period) with offering 0 and getting 0:

$$y - w^* \;\geq\; 0 \qquad (2.3.7)$$

Together with (2.3.5) at its lowest wage:

$$y - e \;\geq\; w_0 + \frac{1-\delta}{\delta(1-p)}\, e.$$

Solve for δ: write $S = y - e - w_0 > 0$ for the surplus of the relationship. Then $S\,\delta(1-p) \geq (1-\delta)e \iff \delta\big[(1-p)S + e\big] \geq e$:

$$\boxed{\delta \;\geq\; \delta_{\min} = \frac{e}{e + (1-p)(y - e - w_0)}}$$

"the familiar restriction that δ must be sufficiently large" (p. 111). Efficiency ($S > 0$) is necessary, not sufficient.

Off the path (p. 111): after any other history the worker never works, so the firm should induce self-employment ($w = 0$ forever), and the worker should shirk and accept only $w \geq w_0$. Note the subgames now begin **within** stages too — after the worker sees the offer.

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5 · C — renegotiation and outside options (Gibbons pp. 111–112)

- **Permanent self-employment is a strange punishment.** Both players prefer to restart the high-wage relationship — the renegotiation problem of L06-04-punishments-price-wars §2. A firm with **many** workers may refuse to renegotiate with one, to protect the equilibrium with the others; firms may refuse to hire workers fired elsewhere — a possible account of low mid-career mobility among large Japanese firms.

- **Outside jobs set w_0.** If fired workers find jobs they prefer to self-employment, the **net wage in those jobs** plays the role of w_0. If a fired worker loses nothing, there is no punishment and no effort in any SPE.
- **Sovereign debt** (Bulow and Rogoff 1989, p. 112): if a defaulting country can replicate long-term loans with cash-in-advance trades, creditors have no punishment and lending collapses. Same logic: no loss, no discipline.

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6 · C — Milgrom & Roberts's Shapiro–Stiglitz model (pp. 250–252)

Notation (M&R's, different from Gibbons's): w the current wage, $\bar{w}$ the wage after being fired (net of search costs), g the gain from cheating, p **the probability that cheating is detected**, N a multiplier for the long-term value of the relationship ($N = 1$ for a one-period job, larger the longer and more certain re-hiring is). The employee cheats iff

$$g \;>\; p\,(w - \bar{w})\,N \qquad (8.1)$$

— gain from cheating against the probability of detection times the income lost by being fired. $w - \bar{w}$ is the **rent**.

6.1 · Worked — M&R's numbers (p. 251)

w = USD 50,000, $\bar{w}$ = USD 40,000, g = USD 1,000, $p = 0.05$, $N = 1$:

$$p(w - \bar{w})N = 0.05 \times 10{,}000 \times 1 = 500 \;<\; 1{,}000 = g \quad \Rightarrow \quad \text{cheating pays.}$$

The efficiency wage — the smallest wage that deters (p. 252):

$$\boxed{w = \bar{w} + \frac{g}{Np}} = 40{,}000 + \frac{1{,}000}{0.05N} = 40{,}000 + \frac{20{,}000}{N} : \qquad N = 1 \to 60{,}000, \quad N = 2 \to 50{,}000.$$

The USD 50,000 job deters cheating only if the relationship is worth twice a single period.

Lord Clive and the Maghribi traders (p. 252) in the formula: the Company's servants had **large** g (power to extort) and **small** p (remote posts), so only a large premium worked; the Maghribi coalition **lowered** $\bar{w}$ and **raised** N (an ostracized agent lost all merchants' business) and **raised** p (dense correspondence) — L06-06-institutions-reputation §3.

6.2 · The two models are one inequality

Gibbons (2.3.5) rearranged	M&R (8.1)	Note
gain from shirking e	g	
detection probability $1 - p$	p	**opposite uses of the letter** p
rent $w^* - e - w_0$	$w - \bar{w}$	so $\bar{w} \leftrightarrow w_0 + e$: M&R's outside wage is already net of the effort an honest worker supplies in either job

Gibbons (2.3.5) rearranged	M&R (8.1)	Note
$\delta/(1-\delta)$	N	Gibbons counts the future only; M&R's $N=1$ for a one-period job counts the current period's rent too (footnote 4) — same algebra, different bookkeeping

Calibrated into Gibbons — $e=1$, $w_0=39$ (so $w_0+e=40=\bar{w}$), $p=0.95$ (detection 0.05), in USD thousands:

$$w^* = 40 + \frac{1-\delta}{\delta} \times \frac{1}{0.05} = 40 + 20\,\frac{1-\delta}{\delta}: \qquad \delta=\tfrac{1}{2}\ (N=1)\to 60, \qquad \delta=\tfrac{2}{3}\ (N=2)\to 50.$$

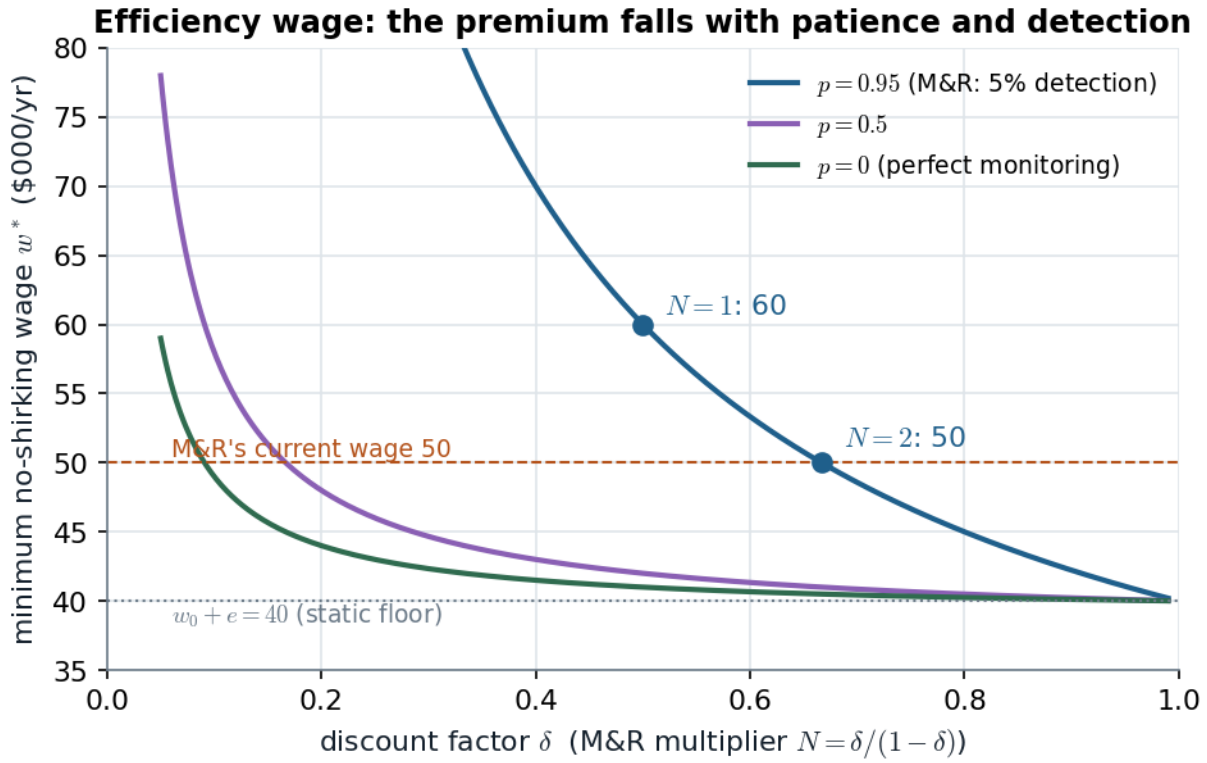


Figure — Gibbons (2.3.5) at M&R's calibration: $w_0+e=40$, $e=1$; detection $1-p$ of 5%, 50% and 100%. Source: Gibbons p. 110; M&R pp. 251–252.

The vertical gap between a curve and the dotted floor is the premium. At 5% detection the premium is twenty times what perfect monitoring needs at the same δ; the M&R wage of 50 crosses the blue curve exactly at $\delta=2/3$, $N=2$.

> **Interactive companion.** The No-Shirking Wage (`companions/companion-efficiency-wage.html`) — move δ, the detection rate and the effort cost and watch w^*, the rent, and the two value lines V_e, V_s cross; raise y to see when the firm will pay. Defaults reproduce M&R's USD 60,000.

7 · U — monitoring, the alternative, and the evidence (M&R pp. 252–255, 260–261)

Monitoring against wages (pp. 254–255). If detection costs $M(p)$ per period, the firm solves the *minimum cost implementation problem*

$$\min_{p,w} \; M(p) + w \quad \text{s.t.} \quad p(w - \bar{w})N \geq g \qquad (8.2)$$

The wage is set at its floor, $w = \bar{w} + g/(Np)$ (8.3); substituting, minimize $M(p) + \bar{w} + g/(Np)$ (8.4). First-order condition $M'(p^*) = g/(p^{*2}N)$ (8.5); second-order $M''(p^*) + 2g/(p^{*3}N) \geq 0$ (8.6). **How does p^* move with N?** Differentiate (8.5):

$$M''(p^*)\,dp^* = -\frac{2g}{p^{*3}N}\,dp^* - \frac{g}{p^{*2}N^2}\,dN \;\Rightarrow\; \frac{dp^*}{dN} = -\frac{g/(p^{*2}N^2)}{M''(p^*) + 2g/(p^{*3}N)} < 0$$

by (8.6) holding strictly. "As the potential length of a relationship increases, there is more trust and less direct monitoring" (p. 255).

Why not explicit incentive pay? (pp. 252–253). Wealth limits (the worker cannot bear low pay or fines), and **subjective evaluations**: a firm that pays on its own assessment is tempted to underrate; a flat efficiency wage removes that temptation. (The incentive-contract side is moral hazard — out of scope.)

Unemployment as the stick (p. 253): not every firm can pay above every other; Shapiro and Stiglitz close the model with unemployment, so a fired worker loses income while searching — at the social cost of idle labour. Okuno-Fujiwara's alternative, lifetime employment with no mid-career hiring, is "very fragile" (p. 254).

MacRents (pp. 260–261, Krueger 1990). Company-owned fast-food outlets, managed by salaried managers who monitor less than franchisee-owners, should pay more. They do: supervisors earn **8.9%** more and full-time non-managerial workers **1.7%** more than in franchised outlets; company outlets are only two thirds as likely to start workers at the minimum wage, and give more generous benefits (free meals).

Links

- Previous: L06-06-institutions-reputation. Next: L06-08-relational-contracts.
- Template: L06-01-recap-grim-trigger §3. Rent and incomplete contracts: rules/06.
- Code: L06-11-python §7. Formulas: L06-09-formulas §1.4, §2.5.
- Exercises: M&R Quantitative Exercises 1 (p. 285) and 3 (p. 286).

Synthesis · A relational contract is a self-enforcing promise

Claim. A *relational contract* — M&R's term for an agreement that leaves decisions to discretion and relies on good faith (p. 259) — is enforced not by a court but by **the value of the relationship itself**. Every case in this lecture is one inequality with four numbers: the per-period value of cooperating, the best one-period gain from cheating, what is left after the breakdown, and δ. Organizational design is the art of moving those four numbers. Prev: L06-07-efficiency-wages · Next: L06-09-formulas · Map: L06-00-map

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1 · C — the four cases side by side

$$\text{cooperation is self-enforcing} \iff \underbrace{v^d - v^c}_{\text{gain from cheating today}} \leq \frac{\delta}{1-\delta}\,\pi_{\text{det}}\ \underbrace{(v^c - v^p)}_{\text{per-period stake}}$$

with $\pi_{\text{det}} = 1$ under perfect monitoring.

Relationship	v^c	v^d	v^p	δ^* (book's numbers)	Note
Cournot cartel, symmetric	$(a-c)^2/8$	$9(a-c)^2/64$	$(a-c)^2/9$	$9/17 = 0.529$	L06-02-cournot-collusion
Cournot cartel, 22/23 split	990 / 1035	1122.25 / 1156	900	0.595 / 0.473	L06-02-cournot-collusion §2
Bertrand cartel, n firms	π_M/n	π_M	0	$1 - 1/n$	L06-03-bertrand-market-structure
Trust game (trusted party)	$V = 1$	$V + G = 2$	0	$G/(V+G) = 1/2$	L06-05-reputation-trust
Guarantor / law merchant	fee 0.1	bond 2	0	0.95	L06-06-institutions-reputation
Efficiency wage (worker)	$w^* - e$	w^* (no effort), caught w.p. $1-p$	w_0	$w^* \geq w_0 + e + \frac{(1-\delta)e}{\delta(1-p)}$	L06-07-efficiency-wages

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2 · C — the design levers

Lever	Moves	Example in this lecture
Raise the stake $v^c - v^p$	denominator up	efficiency-wage rent; a brand's resale value; a harsher (Abreu) punishment

Lever	Moves	Example in this lecture
Cut the temptation $v^d - v^c$	numerator down	most-favoured-customer clauses; a bond that is lost on cheating; sharing a cartel so no member is "worse treated"
Raise δ	patience up	frequent interaction (f), stable market (low h), growth (g) — Cabral (9.5); lifetime employment
Raise detection	π_{det} up	monitoring $M(p)$; credit bureaus; the Maghribi correspondence; transparent prices
Make judging easy	removes ambiguity	corporate culture's simple rules; private judges
Remove the end	no terminal period	sale of the business, brand names that outlive owners

Every lever has a social sign. The same levers that let a firm buy honesty from its workers let competitors keep prices high — which is why antitrust attacks transparency devices (GE's price book, Cabral Box 9.5) that a personnel manager would welcome.

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3 · C — when relational contracts fail

1. **A known last period** — the end game (M&R p. 266) and finite-horizon unravelling (Lecture 5).
2. **Too little patience** — $\delta < \delta^*$; a financially weak firm whose δ falls starts the war (Cabral p. 226).
3. **Nothing to lose** — a party with zero stake (the high-cost firm in an efficient cartel; a worker whose next job is as good) cannot be disciplined at any δ.
4. **Noisy monitoring** — punishments must fire on bad luck too: price wars on the path (Green–Porter).
5. **Renegotiation** — a punishment both parties would rather skip is not credible.
6. **Ambiguity** — when nobody can tell whether the right thing was done, there is nothing to condition the punishment on.

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4 · U — where this sits in the course

Relational contracts are the bridge between the **complete-information repeated games** of Lectures 5–6 and the **incomplete contracts** of M&R Ch. 5: a contract can be incomplete (courts cannot verify effort, quality or honour) and still be enforced, because the parties themselves can see enough to punish. Lectures 7–8 drop complete information: then reputation becomes a **belief about a type** (Tadelis Ch. 17, background only) rather than a strategy profile.

Links

- Map L06-00-map · formulas L06-09-formulas · pitfalls L06-10-pitfalls.
- Rules: rules/06 ("Relational and implicit contracts"), rules/game-theory.
- Next lectures: Lecture 7, Lecture 8.

Lecture 6 · Formulas — every result, every number, the study guide mapped

Prev: L06-08-relational-contracts · Next: L06-10-pitfalls · Map: L06-00-map

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0 · The exam specification, mapped

No slides yet: the specification is the Lecture 6 block of `Map/study-guide.md`, item by item.

#	Study-guide item	Note
1	Tadelis §10.4 tacit collusion (p. 201)	L06-02-cournot-collusion §2, §6
2	Tadelis §10.5 reputation (pp. 204–209)	L06-05-reputation-trust §1; L06-06-institutions-reputation
3	Gibbons §2.3.C Cournot collusion (p. 102)	L06-02-cournot-collusion §3–§5; L06-04-punishments-price-wars §1, §3–§4
4	Gibbons §2.3.D efficiency wages (p. 107)	L06-07-efficiency-wages §2–§5
5	M&R "Reputations as contract enforcers" (p. 259), the repeated trust game (pp. 261–263)	L06-05-reputation-trust §2–§4
6	M&R "live and let live" (p. 264), limits of reputation (p. 265)	L06-05-reputation-trust §5–§6
7	Formal: MWG §12.D (p. 400)	L06-03-bertrand-market-structure §1 (Props. 12.D.1–12.D.2)
8	Cabral Ch. 9: stability (p. 218), price wars (p. 222)	L06-03-bertrand-market-structure; L06-04-punishments-price-wars §2–§5
9	M&R efficiency wages (p. 250)	L06-07-efficiency-wages §6–§7
—	Recap needed from Lecture 5 (one-stage deviation, grim trigger)	L06-01-recap-grim-trigger
—	The unifying inequality	L06-08-relational-contracts

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1 · Every definition and result

1.1 · The machinery (Tadelis)

Result	Statement	Where derived
Def. 10.2–10.3 (pp. 193–194)	PV $\sum \delta^{t-1} v^t$; average $(1-\delta)\sum \delta^{t-1} v^t$	L06-01-recap-grim-trigger §2

Result	Statement	Where derived
Prop. 10.2 (p. 196)	stage Nash after every history is an SPE	§2
Prop. 10.3 (p. 198)	one-stage deviation principle for infinitely repeated games	§2
(10.3), p. 200	PD: $4 + \frac{4\delta}{1-\delta} \geq 5 + \frac{\delta}{1-\delta} \iff \delta \geq \frac{1}{4}$	§3.1
(10.5), p. 202	$\delta \geq \dfrac{v_i^d - v_i^c}{v_i^d - v_i^*}$	§3
equivalent form	$v^d - v^c \leq \frac{\delta}{1-\delta}(v^c - v^p)$; average form $v^c \geq (1-\delta)v^d + \delta v^p$	§3

1.2 · Collusion

Result	Statement	Where
(10.4), p. 202	$BR_i(q_j) = (90 - q_j)/2$	L06-02-cournot-collusion §2
(2.3.2), p. 103	$\frac{1}{1-\delta}\frac{\pi_m}{2} \geq \pi_d + \frac{\delta}{1-\delta}\pi_c \iff \delta \geq \frac{9}{17}$	§3
Gibbons p. 104	$q^*(\delta) = \dfrac{9 - 5\delta}{3(9-\delta)}(a-c)$ for $\delta < 9/17$	§4
n-firm Cournot	$\delta^* = \dfrac{(n+1)^2}{n^2 + 6n + 1}$	§5
(9.1)–(9.4), p. 219	$\frac{\frac{1}{2}\pi_M}{1-\delta} \geq \pi_M \iff \delta \geq \frac{1}{2}$	L06-03-bertrand-market-structure §1
MWG Prop. 12.D.1 (p. 401)	Nash reversion is an SPE of repeated Bertrand iff $\delta \geq \frac{1}{2}$	§1
MWG Prop. 12.D.2 (p. 403)	$\delta \geq \frac{1}{2}$: any $p \in [c, p_M]$ sustainable; $\delta < \frac{1}{2}$: every SPE prices at c	§1
n-firm Bertrand	$\delta \geq 1 - 1/n$ (MWG Ex. 12.D.2, p. 405; Cabral p. 227)	§2
(9.5), p. 220	$\delta = \dfrac{(1+g)(1-h)}{1 + r/f}$; up in f, g; down in h, r	§3
(2.3.3), p. 105	$\delta(\frac{\pi_m}{2} - \pi(x)) \geq \pi_d - \frac{\pi_m}{2}$	L06-04-punishments-price-wars §1
(2.3.4), p. 106	$\delta(\frac{\pi_m}{2} - \pi(x)) \geq \pi_{dp}(x) - \pi(x)$	§1

1.3 · Reputation

Result	Statement	Where
Trust game, p. 205	$\frac{1}{1-\delta} \geq 2 \iff \delta \geq \frac{1}{2}$	L06-05-reputation-trust §1
M&R footnote 11, p. 263	$N = \frac{1}{(1+r)^t - 1} + 1 = \frac{1}{1-\delta}$, $\delta = (1+r)^{-t}$	§2

Result	Statement	Where
M&R trust box, pp. 261–263	exact: $NV \geq V + G \iff \delta \geq \frac{G}{V+G}$ (M&R print $NV > G$)	§2
Guarantor, p. 207	$\frac{0.1}{1-\delta} \geq 2 \iff \delta \geq 0.95$	L06-06-institutions-reputation §1

1.4 · Efficiency wages

Result	Statement	Where
Gibbons p. 109	$V_e = \frac{w^* - e}{1-\delta}$	L06-07-efficiency-wages §3
Gibbons p. 110	$V_s = \frac{(1-\delta)w^* + \delta(1-p)w_0}{(1-\delta p)(1-\delta)}$	§3
(2.3.5), p. 110	$w^* \geq w_0 + e + \frac{(1-\delta)e}{\delta(1-p)}$	§3
(2.3.6), p. 110	$p = 0$: $w^* \geq w_0 + \left(1 + \frac{1-\delta}{\delta}\right)e$	§3
(2.3.7), p. 110	$y - w^* \geq 0$	§4
combined	$\delta \geq \frac{e}{e + (1-p)(y-e-w_0)}$	§4
(8.1), p. 251	cheat iff $g > p(w - \bar{w})N$ ($p =$ **detection**)	§6
p. 252 / (8.3)	efficiency wage $w = \bar{w} + g/(Np)$	§6
(8.2)–(8.6), p. 254	$\min M(p) + w$ s.t. IC; $M'(p^*) = g/(p^{*2}N)$; $dp^*/dN < 0$	§7

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2 · Every number

2.1 · Tadelis's Cournot cartel ($p = 100 - q$, $c = 10$)

	Firm 1 ($q^c = 22$)	Firm 2 ($q^c = 23$)
v^c	990	1035
q^d	33.5	34
v^d	1122.25	**1156** (book prints 1122)
δ^*	**0.595**	**0.473** (book prints 0.392)

Cournot 30 each, price 40, profit 900; monopoly split 22.5 each, price **55** (book's p. 203 "45" is total output), profit 1012.5.

2.2 · Gibbons's symmetric cartel ($a - c = 1$; Tadelis numbers in brackets)

$\pi_m/2 = 1/8$ [1012.5], $\pi_d = 9/64$ [1139.06, $q_d = 33.75$], $\pi_c = 1/9$ [900] $\to \delta^* = 9/17 = 0.529$. At $\delta = 1/2$: $q^* = 6.5/25.5 = 0.2549$ [22.94 each, price 54.12, profit 1012.1]. n-firm cutoffs: 0.529, 0.571, 0.610, 0.643 for $n = 2, 3, 4, 5$; 0.752 at $n = 10$.

2.3 · Bertrand and Cabral

$\delta^* = 0.5,\ 0.667,\ 0.75,\ 0.8,\ 0.9$ for $n = 2, 3, 4, 5, 10$. Cabral p. 221: $r = 10\%$, monthly prices $\to$ $\delta = 0.9917$, $n \leq 121$; annual prices $\to \delta = 0.909$, $n \leq 11$. Box 9.1: industrial diamonds USD 3 $\to$ under

USD 1.80 per carat (1981). Box 9.3: average route contact 402.3. Cement: +1% GDP → −0.5% to −1% relative price.

2.4 · Punishments, reputation, institutions

Abreu at $\delta = 1/2$: (2.3.3) holds for $x \notin (1/8, 3/8)$, (2.3.4) for $x \in [3/10, 1/2] \to x/(a-c) \in [3/8, 1/2]$. Trust game $\delta \geq 1/2$; guarantor fee 0.1, bond 2 → $\delta \geq 0.95$ (bond need only exceed 1). M&R multiplier at $r = 10\%$: $N = 11$ (yearly), 2.64 (every 5 years), 1.17 (every 20 years).

2.5 · Efficiency wages

M&R: $w = 50$, $\bar{w} = 40$, $g = 1$, $p = 0.05$ (USD thousands) → $0.05 \times 10 \times N$ vs 1: cheat at $N = 1$; efficiency wage $40 + 20/N = 60$ ($N = 1$), 50 ($N = 2$). In Gibbons's terms ($w_0 = 39$, $e = 1$, $p = 0.95$): $w^* = 40 + 20(1-\delta)/\delta$ — 60 at $\delta = 1/2$, 50 at $\delta = 2/3$. MacRents: supervisors +8.9%, non-managerial +1.7% in company outlets.

Links

L06-00-map · L06-10-pitfalls · L06-11-python (every number here is asserted there).

Lecture 6 · Pitfalls and a True/False self-test

Prev: L06-09-formulas · Next: L06-11-python · Map: L06-00-map

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Part I — the confusions this material produces

1 · "Any deviation" instead of the best one-period deviation

The cutoff (10.5) uses v^d, the **most tempting** deviation — the stage best response to the others' cooperative play. A firm in Tadelis's cartel that deviates from 22 to 30 gains $(100-30-23-10)\times 30 = 1110$; to 33.5 it gains 1122.25. Using 30 gives $\delta^* = (1110-990)/(1110-900) = 0.571$ instead of 0.595 — an understated requirement. → L06-02-cournot-collusion §2.

2 · One cutoff for an asymmetric cartel

Each player has his own cutoff; the cartel survives iff $\delta \geq \max_i \delta_i^*$. In the 22/23 split the **smaller** firm binds (0.595 against 0.473). And copy the book with care: its 0.392 for firm 2 plugs in the wrong rival quantity.

3 · "Too little patience means back to the stage game" — true for Bertrand, false for Cournot

Bertrand: the condition $\frac{\pi(p)/n}{1-\delta} \geq \pi(p)$ does not depend on p, so below $1-1/n$ nothing above cost survives (MWG Prop. 12.D.2). Cournot: below 9/17 the cartel **shrinks** to $q^*(\delta)$ and still beats Cournot. → L06-03-bertrand-market-structure §1.

4 · "Price competition makes collusion harder" — not at $n = 2$

Bertrand needs 1/2, Cournot $9/17 > 1/2$ with two firms; from $n = 3$ the ranking flips ($0.667 > 0.571$). Harsher stick against bigger temptation.

5 · Reading a price war as proof that someone cheated

Under imperfect monitoring (Green–Porter) wars happen **on** the equilibrium path after bad demand shocks, and nobody cheats. Under observed shocks (Rotemberg–Saloner) prices fall in **booms**. → L06-04-punishments-price-wars §3–§4.

6 · Grim trigger as "the" punishment

It is the simplest, not the strongest: Abreu's one-period stick sustains the monopoly outcome at $\delta = 1/2 < 9/17$. And "forever" invites renegotiation.

7 · Forgetting the off-path half of a strategy

"Cooperate as long as player 1 trusts me" is a Nash equilibrium, not an SPE (Tadelis footnote 9). A repeated-game strategy specifies play after **every** history.

8 · Mixing the two p's

Gibbons's p = probability that **shirking goes undetected** (output still high); M&R's p = probability that **cheating is detected**. Gibbons's premium has $1-p$ in the denominator, M&R's has p. Same economics, opposite letters. → L06-07-efficiency-wages §6.2.

9 · M&R's $NV > G$ taken literally

From the box, cheating yields $V+G$ in the round of cheating; the exact comparison is $NV \geq V+G \iff \delta \geq G/(V+G)$. With $V = G = 1$, $NV > G$ would give $N > 1$, i.e. any $\delta > 0$ — wrong; the exact form gives $\delta \geq 1/2$.

10 · "Efficient, therefore sustainable"

$y - e > w_0$ makes the employment relation efficient; it is sustainable only if $y \geq w^*$, i.e. $\delta \geq e/(e + (1-p)(y-e-w_0))$. With M&R's calibration and $y = 41$, that is $\delta \geq 0.952$.

11 · Reputation without observability or judgment

Reputation needs future partners to **see** the past and to **judge** it. Remove either (no communication, ambiguous duties) and the incentive disappears — the case for law merchants, credit bureaus and corporate culture. → L06-05-reputation-trust §6.

12 · Reproducing a finite-horizon argument in an infinite game, or vice versa

Known last period: unravels (end game, CEO near retirement). Indefinite continuation with probability δ: cooperation possible. The question always states which one.

Part II — True/False (original items)

Scoring as in `macro-1`*:* $\max\{0,\ \textit{correct} - \textit{wrong}/2\} \times$ *item value; blanks do not count as wrong.*

1. In the infinitely repeated Cournot duopoly with $P = a - Q$ and constant marginal cost, grim trigger sustains half the monopoly output for every player iff $\delta \geq 9/17$. **True.** $\pi_m/2 = (a-c)^2/8$, $\pi_d = 9(a-c)^2/64$, $\pi_c = (a-c)^2/9$; (10.5) gives $(1/64)/(17/576) = 9/17$; the symmetric game gives both firms the same cutoff.

2. If $\delta < 9/17$ in that duopoly, the only subgame-perfect outcome of the repeated game under Nash-reversion strategies is the Cournot outcome in every period. **False.** Grim trigger still sustains any symmetric $q \geq q^*(\delta) = \frac{9-5\delta}{3(9-\delta)}(a-c)$, strictly below Cournot for $\delta > 0$ — e.g. $0.2549(a-c)$ at $\delta = 1/2$.

3. In the repeated Bertrand duopoly with grim strategies, the cutoff discount factor depends on the collusive price chosen. **False.** Conform $\frac{\frac{1}{2}\pi(p)}{1-\delta}$ versus deviate $\pi(p)$: $\pi(p)$ cancels and the cutoff is $1/2$ for every $p \in (c, p_M]$ (MWG Prop. 12.D.2).

4. With n symmetric Bertrand firms, collusion at the monopoly price is sustainable iff $\delta \geq 1 - 1/n$. **True.** $\frac{\pi_M/n}{1-\delta} \geq \pi_M \iff 1/n \geq 1-\delta$.

5. With two firms, tacit collusion requires more patience under Bertrand than under Cournot competition. **False.** $1/2 < 9/17$: with two firms Bertrand needs less; the ranking reverses from three firms on.

6. In Tadelis's 22/23 split of the monopoly output, the firm producing 22 needs a higher discount factor than the firm producing 23. **True.** It gains more by deviating (132.25 vs 121) and loses less per period from Cournot (90 vs 135): 0.595 against 0.473.

7. By Cabral's (9.5), collusion is easier in an industry that reprices daily than in one that reprices once a year, other things equal. **True.** $\delta = \frac{(1+g)(1-h)}{1+r/f}$ rises with the frequency f.

8. An increase in the hazard that the industry disappears makes collusion easier because firms have less to lose from a price war. **False.** A higher h lowers δ: the future punishment matters less, so the temptation wins more often — collusion is **harder**.

9. In Cabral's model, if two firms meet in two identical markets, multimarket contact halves the discount factor needed to collude. **False.** Both the gain from deviating and the punishment double; the cutoff stays 1/2. Contact helps only when the markets differ.

10. In a Green–Porter cartel with unobserved demand shocks, price wars occur on the equilibrium path even though no firm ever deviates. **True.** Low prices must trigger punishment or every firm would cut secretly; bad shocks then start wars with nobody cheating.

11. If demand shocks are observed before pricing and are independent over time, a collusive cartel's price tends to be lowest when demand is highest. **True.** The deviation gain scales with current demand while the future loss does not; the constraint binds in booms, so the cartel cuts its price there (Rotemberg–Saloner).

12. In Abreu's two-phase strategy at $\delta = 1/2$, any punishment output x larger than the Cournot quantity $(a-c)/3$ sustains the monopoly outcome. **False.** The collusive-phase constraint needs $x \geq \frac{3}{8}(a-c)$, and $\frac{1}{3} < x < \frac{3}{8}$ fails it; the feasible band is $x/(a-c) \in [3/8, 1/2]$.

13. A punishment that hurts the punisher as well as the deviator can never be part of a subgame-perfect equilibrium. **False.** Both grim trigger and Abreu's stick hurt the punisher; they are SPE because carrying out the punishment is a best response given what follows. The issue they raise is renegotiation, not subgame perfection.

14. In Tadelis's repeated trust game with payoffs $(1,1)$, $(-1,2)$ and $(0,0)$, cooperation is sustainable by grim trigger iff $\delta \geq 1/2$. **True.** $\frac{1}{1-\delta} \geq 2$.

15. In M&R's trust box, the decision maker's exact no-cheating condition is $NV \geq G$. **False.** Cheating gives $V + G$ in the round of cheating; the exact condition is $NV \geq V + G$, i.e. $\delta \geq G/(V+G)$.

16. Reputation can discipline a decision maker even if he never meets the same offeror twice, provided each offeror observes how he treated earlier offerors. **True.** M&R p. 263: the calculation is the same as with one offeror; that is what a reputation is.

17. A guarantor who earns a fee of 0.1 per period and could pocket a bond of 2 keeps the contract iff $\delta \geq 0.95$. **True.** $\frac{0.1}{1-\delta} \geq 2 \iff \delta \geq 0.95$.

18. In Gibbons's efficiency-wage model, the wage premium needed to induce effort falls as the probability that a shirker still produces high output rises. **False.** The premium $\frac{(1-\delta)e}{\delta(1-p)}$ **rises** with p: shirking is caught less often, so a larger rent is needed.

19. If shirking is always detected ($p = 0$ in Gibbons), the firm can induce effort by paying exactly $w_0 + e$. **False.** It must still pay $w_0 + e/\delta$: the gain e today must be covered by the rent from tomorrow, which is discounted. Only as $\delta \to 1$ does $w^* \to w_0 + e$.

20. In the efficiency-wage model, $y - e > w_0$ guarantees that the high-wage, high-output equilibrium exists for every δ. **False.** It also needs $y \geq w^*$, i.e. $\delta \geq e/(e + (1-p)(y - e - w_0))$.

21. With M&R's numbers (rent USD 10,000, detection 5%, gain USD 1,000), a worker with $N = 1$ prefers to cheat. **True.** Expected loss $0.05 \times 10{,}000 = 500 < 1{,}000$; the efficiency wage would be USD 60,000.

22. A longer expected relationship (higher N) leads, in M&R's minimum-cost implementation problem, to more intensive monitoring. **False.** $dp^*/dN < 0$: a longer relationship substitutes for monitoring ("more trust and less direct monitoring", p. 255).

23. A known final period weakens reputational incentives as it approaches. **True.** The end-game problem (M&R p. 266): at the last date there is no future to protect, and the argument creeps backwards; hence more explicit incentive pay for CEOs near retirement.

24. A firm's brand name can sustain cooperation by short-lived owners because the brand's resale value is lost if trust is abused under it. **True.** Kreps's brand acts as a bond (Tadelis pp. 207–208): the resale value above 1 exceeds the one-period gain from defecting.

Links

L06-00-map · L06-09-formulas · L06-11-python · style: `rules/00_estilo_avaliacao.md`.

Lecture 6 · Computation in Python

> Every routine the lecture needs, each with `assert` self-checks against the books' own numbers. numpy only; the blocks run top to bottom in one session (they share names). Prev: L06-10-pitfalls · Map: L06-00-map · Formulas: L06-09-formulas

Reuse. The project has no repeated-game code yet (`Listas/*-check.py` and `Aula/**/check_*.py` cover Lectures 1–2 and the Aula 1–2 notes). The figure script `figures/make_figures.py` and the companion check `companions/check_companions.py` re-implement the same expressions independently on purpose: they are the second path the numbers are verified on.

---

1 · The one inequality → L06-01-recap-grim-trigger

```
import numpy as np

def discounted(stream, delta):
    """Present value of a finite payoff stream (a long horizon stands in for
    ↪ infinity)."""
    stream = np.asarray(stream, float)
    return float(np.sum(stream * delta ** np.arange(len(stream))))

def grim_cutoff(v_coop, v_dev, v_pun):
    """delta* = (v^d - v^c) / (v^d - v^p): Tadelis (10.5), p. 202.

    v_coop: per-period payoff on the cooperative path; v_dev: best one-period
 ↪  deviation
    payoff; v_pun: per-period payoff in the punishment (stage Nash) forever after.
    Requires v_dev > v_coop > v_pun; returns the smallest delta that deters the
 ↪  deviation.
    """
    assert v_dev > v_coop > v_pun, "not a temptation/punishment configuration"
    return (v_dev - v_coop) / (v_dev - v_pun)

# Tadelis's repeated Prisoner's Dilemma (p. 197): mum 4, fink against mum 5,
 ↪  fink-fink 1.
assert grim_cutoff(4, 5, 1) == 0.25                                  # (10.3), p. 200
for d in (0.24, 0.25, 0.26):                                          # one-shot
 ↪  deviation at t = 0
    T = 4000
    conform = discounted([4] * T, d)
    deviate = discounted([5] + [1] * (T - 1), d)
    assert (conform >= deviate - 1e-9) == (d >= 0.25)
```

---

2 · Tadelis's tacit collusion, with the corrected firm-2 numbers → L06-02-cournot-collusion

```
A, C = 100.0, 10.0                    # p = 100 - q, c_i(q_i) = 10 q_i  (p. 201)
```

```
def profit(qi, qj, a=A, c=C):
    return (a - qi - qj - c) * qi

def br(qj, a=A, c=C):
    return (a - c - qj) / 2           # (10.4), p. 202

assert profit(30, 30) == 900 and profit(22.5, 22.5) == 1012.5                    # p.
 ↪ 201
assert profit(22, 23) == 990 and profit(23, 22) == 1035                          # p.
 ↪ 201
assert br(23) == 33.5 and br(22) == 34                                            # p.
 ↪ 202
assert profit(33.5, 23) == 1122.25                                                #
 ↪ firm 1, p. 202
assert profit(34, 22) == 1156          # firm 2 deviates against q1 = 22; the book
 ↪ prints 1122
assert (100 - 23 - 34) * 34 - 10 * 34 == 1122   # ...which is what using q_j = 23
 ↪ gives
d1 = grim_cutoff(990, 1122.25, 900)
d2 = grim_cutoff(1035, 1156, 900)
assert round(d1, 3) == 0.595                     # p. 202, unchanged
assert abs(d2 - 121 / 256) < 1e-12               # 0.473, not the printed 0.392
assert round(grim_cutoff(1035, 1122, 900), 3) == 0.392
assert d1 > d2                                   # the book's conclusion survives the
 ↪ fix
assert A - 45 == 55                              # collusive price is 55 (p. 203
 ↪ prints 45)
```

---

3 · The symmetric n-firm Cournot cartel → L06-02-cournot-collusion

```
def cournot_cartel(a, c, n):
    """Per-firm profits: collusive share pi_m/n, best deviation pi_d, Cournot pi_N.

    Linear demand P = a - Q, constant marginal cost c < a, n identical firms.
    """
    s = a - c
    pi_share = s ** 2 / (4 * n)
    q_dev = s * (n + 1) / (4 * n)                  # best response to (n-1) * s/(2n)
    pi_dev = q_dev ** 2
    pi_N = s ** 2 / (n + 1) ** 2
    return pi_share, pi_dev, pi_N

def cournot_threshold(n):
    """Closed form of grim_cutoff for the n-firm cartel: (n+1)^2 / (n^2 + 6n + 1)."""
    return (n + 1) ** 2 / (n ** 2 + 6 * n + 1)

def best_sustainable_q(a, c, n, delta):
    """Lowest symmetric per-firm output that grim trigger (Nash reversion) sustains.
```

```
    Condition: (a - nq - c) q >= (1 - delta) pi_d(q) + delta pi_N, pi_d(q) = (a - c -
↪   (n-1) q)^2 / 4.
    The condition is a concave quadratic in q that vanishes at the Cournot output, so
↪   the
    other root is (product of roots) / q_N. Returns max(q_m / n, that root).
    """
    s = a - c
    c2 = -n - (1 - delta) * (n - 1) ** 2 / 4
    c0 = -(1 - delta) * s ** 2 / 4 - delta * s ** 2 / (n + 1) ** 2
    qN = s / (n + 1)
    return max(s / (2 * n), (c0 / c2) / qN)

# Gibbons p. 103: n = 2 gives delta >= 9/17.
sh, dv, nash = cournot_cartel(1, 0, 2)
assert (sh, dv, nash) == (1 / 8, 9 / 64, 1 / 9)
assert abs(grim_cutoff(sh, dv, nash) - 9 / 17) < 1e-12
for n in range(2, 30):
    assert abs(grim_cutoff(*cournot_cartel(100, 10, n)) - cournot_threshold(n)) <
↪   1e-12
# Gibbons p. 104: q* = (9 - 5 delta) / (3 (9 - delta)) (a - c) for delta < 9/17.
for d in np.linspace(0, 9 / 17, 50):
    assert abs(best_sustainable_q(1, 0, 2, d) - (9 - 5 * d) / (3 * (9 - d))) < 1e-12
assert best_sustainable_q(1, 0, 2, 0.0) == 1 / 3 and best_sustainable_q(1, 0, 2, 0.6)
↪   == 0.25
# brute force: q* is the smallest q on a fine grid that satisfies the condition (n =
↪   4)
a, c, n, d = 100, 10, 4, 0.4
q = np.linspace(90 / 8, 90 / 5, 200001)
ok = (a - n * q - c) * q >= (1 - d) * (a - c - (n - 1) * q) ** 2 / 4 + d * (90 / 5)
↪   ** 2 - 1e-9
assert abs(q[ok].min() - best_sustainable_q(a, c, n, d)) < 1e-3
# Tadelis's numbers at Gibbons's delta = 1/2: q* = 22.94 per firm, price 54.12
assert round(best_sustainable_q(100, 10, 2, 0.5), 2) == 22.94
```

---

4 · Bertrand, the number of firms, and Cabral's discount factor → L06-03-bertrand-market-structure

```
def bertrand_threshold(n):
    """Nash reversion sustains any p in (c, p_m] iff delta >= 1 - 1/n (MWG Prop.
↪   12.D.1 for n = 2)."""
    return 1 - 1 / n

def cabral_delta(r, f=1, h=0.0, g=0.0):
    """Cabral (9.5), p. 220: delta = (1 + g)(1 - h) / (1 + r/f)."""
    return (1 + g) * (1 - h) / (1 + r / f)

assert bertrand_threshold(2) == 0.5                                    # Cabral (9.4),
↪   p. 219
```

```
# collusion holds iff share/(1 - delta) >= whole monopoly profit
for n in (2, 3, 10):
    for d in (bertrand_threshold(n) - 1e-6, bertrand_threshold(n) + 1e-6):
        assert ((1 / n) / (1 - d) >= 1) == (d >= bertrand_threshold(n))
# Cabral p. 221: 10% a year, monthly prices -> "dozens" of firms can collude
d = cabral_delta(0.10, f=12)
assert round(d, 4) == 0.9917 and int(np.floor(1 / (1 - d) + 1e-9)) == 121
assert cabral_delta(0.10, f=1) < cabral_delta(0.10, f=12)          # delta rises with
 ↪ f
assert cabral_delta(0.10, 12, h=0.05) < d < cabral_delta(0.10, 12, g=0.02)
# Cabral p. 229: two identical markets -> the same threshold, 1/2
d = 0.5
assert np.isclose(2 * 0.5 / (1 - d), 2 * 1.0)
```

---

5 · Abreu's carrot and stick → L06-04-punishments-price-wars

```
def two_phase_band(delta):
    """Punishment outputs x/(a - c) that satisfy Gibbons (2.3.3) and (2.3.4) (pp.
     ↪ 105-106).

    (2.3.3): delta (1/8 - pi(x)) >= 9/64 - 1/8      (collusive phase)
    (2.3.4): delta (1/8 - pi(x)) >= pi_dp(x) - pi(x) (punishment phase)
    with pi(x) = (1 - 2x) x and pi_dp(x) = (1 - x)^2 / 4. Grid search on [0, 1/2].
    """
    x = np.linspace(0, 0.5, 500001)
    pix = (1 - 2 * x) * x
    ok = (delta * (1 / 8 - pix) >= 1 / 64 - 1e-12) & \
         (delta * (1 / 8 - pix) >= (1 - x) ** 2 / 4 - pix - 1e-12)
    return x[ok].min(), x[ok].max()

lo, hi = two_phase_band(0.5)
assert abs(lo - 3 / 8) < 1e-5 and abs(hi - 1 / 2) < 1e-5            # Gibbons p. 106
assert 0.5 < 9 / 17          # the carrot and stick reaches the monopoly outcome grim
 ↪ trigger cannot
```

---

6 · Reputation: trust game, M&R's multiplier, the guarantor → L06-05-reputation-trust, L06-06-institutions-reputation

```
def trust_threshold(V, G):
    """Honour trust iff V/(1 - delta) >= V + G, i.e. delta >= G / (V + G)."""
    return G / (V + G)

def mr_multiplier(r, t=1):
    """M&R footnote 11, p. 263: N = 1/[(1 + r)^t - 1] + 1 = 1/(1 - delta), delta = (1
     ↪ + r)^-t."""
    return 1 / ((1 + r) ** t - 1) + 1
```

```
assert trust_threshold(1, 1) == 0.5            # Tadelis Fig. 10.1 payoffs: delta >=
 ↪ 1/2, p. 205
r = 0.05
assert np.isclose(mr_multiplier(r), 1 / (1 - 1 / (1 + r)))
# M&R's honour condition N V >= V + G is the same inequality as delta >= G/(V + G)
for V, G, d in [(1, 1, 0.49), (1, 1, 0.51), (3, 1, 0.24), (3, 1, 0.26)]:
    assert (V / (1 - d) >= V + G) == (d >= trust_threshold(V, G))
# Law merchant (Tadelis p. 207): fee 0.1 forever vs keeping the bond 2 once
assert grim_cutoff(0.1, 2.0, 0.0) == 0.95
```

---

7 · Efficiency wages → L06-07-efficiency-wages

```
def w_star(w0, e, p, delta):
    """Gibbons (2.3.5), p. 110: lowest wage at which effort beats shirking.

    w0: self-employment wage; e: disutility of effort; p: probability that output is
 ↪  high
    when the worker shirks (so 1 - p is the detection probability); delta in (0, 1).
    """
    return w0 + e + (1 - delta) * e / (delta * (1 - p))

def worker_values(w, w0, e, p, delta):
    """Present values of effort and of shirking at wage w (Gibbons pp. 109-110)."""
    Ve = (w - e) / (1 - delta)
    Vs = ((1 - delta) * w + delta * (1 - p) * w0) / ((1 - delta * p) * (1 - delta))
    return Ve, Vs

def firm_delta_min(y, w0, e, p):
    """Smallest delta with y >= w*: (2.3.5) and (2.3.7) together (Gibbons p. 110)."""
    surplus = y - e - w0
    return e / (e + (1 - p) * surplus) if surplus > 0 else None

def mr_efficiency_wage(w_bar, g, P, N):
    """M&R p. 252: smallest wage deterring cheating, w_bar + g / (N P)."""
    return w_bar + g / (N * P)

# the two values coincide exactly at w*
for args in [(39, 1, 0.95, 0.5), (10, 2, 0.3, 0.8), (0, 5, 0.0, 0.2)]:
    w = w_star(*args)
    Ve, Vs = worker_values(w, *args)
    assert np.isclose(Ve, Vs)
    assert worker_values(w + 0.01, *args)[0] > worker_values(w + 0.01, *args)[1]
# p = 0 reproduces (2.3.6), p. 110: w* = w0 + (1 + (1 - delta)/delta) e
assert np.isclose(w_star(10, 2, 0.0, 0.5), 10 + (1 + 1) * 2)     # w0 + (1 + (1-d)/d)
 ↪ e
# M&R p. 251: rent 10, detection .05, gain 1 ($000): expected loss .5 < 1 -> cheat at
 ↪ N = 1
assert 0.05 * (50 - 40) * 1 < 1
```

```
assert mr_efficiency_wage(40, 1, 0.05, 1) == 60 and mr_efficiency_wage(40, 1, 0.05,
 ↪  2) == 50
# mapping: g = e, P = 1 - p, N = delta / (1 - delta), w_bar = w0 + e
for d in (0.5, 2 / 3, 0.9):
    assert np.isclose(w_star(39, 1, 0.95, d), mr_efficiency_wage(40, 1, 0.05, d / (1
     ↪  - d)))
# the firm's side at the companion default y = 41
dmin = firm_delta_min(41, 39, 1, 0.95)
assert np.isclose(dmin, 1 / 1.05) and np.isclose(w_star(39, 1, 0.95, dmin), 41)
```

---

8 · Run it

Copy the blocks into one file or a notebook cell in order; every `assert` passes. The figures (`python Notas/L06/figures/make_figures.py`) and the companion check (`python Notas/L06/companions/check_compani` are independent re-implementations of §2–§7.

Links

- Map L06-00-map · formulas L06-09-formulas · pitfalls L06-10-pitfalls.
- The repeated-game basics these routines build on: Lecture 5.
- Rules: rules/game-theory (item 4: patience and observable deviations), rules/06.